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Chebyshev collocation spectral lattice Boltzmann method in generalized curvilinear coordinates

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ABSTRACT

In this work, the Chebyshev collocation spectral lattice Boltzmann method is implemented in the generalized curvilinear coordinates to provide an accurate and efficient low-speed LB-based flow solver to be capable of handling curved geometries with non-uniform grids. The low-speed form of the D2Q9 and D3Q19 lattice Boltzmann equations is transformed into the generalized curvilinear coordinates and then the spatial derivatives in the resulting equations are discretized by using the Chebyshev collocation spectral method and the temporal term is discretized with the fourth-order Runge–Kutta scheme to provide an accurate and efficient low-speed flow solver. All boundary conditions are implemented based on the solution of the governing equations in the generalized curvilinear coordinates. The accuracy and robustness of the solution methodology presented are demonstrated by computing different benchmark and practical low-speed flow problems that are 2D Couette flow between concentric moving cylinders, 2D flow in a gradual expansion duct, 2D regularized trapezoidal cavity flow, and 3D flow in curved ducts of rectangular cross-sections. Results obtained for these test cases are in good agreement with the existing analytical and numerical results. The computational efficiency of the proposed solution methodology based on the Chebyshev collocation spectral lattice Boltzmann method implemented in the generalized curvilinear coordinates is also examined by comparison with the developed second-order finite-difference lattice Boltzmann method that indicates the proposed method provides more accurate and efficient solutions in terms of the CPU time and memory usage. The study shows the present solution methodology is robust and accurate for solving 2D and 3D low-speed flows over practical geometries. Indications are that the solution algorithm based on the CCSLBM in the generalized curvilinear coordinates does not need any filtering or numerical dissipation for stability considerations and thus high accuracy solutions obtained by applying the CCSLBM can be used as benchmark solutions for the evaluation of other LBM-based flow solvers.

1. Introduction

In the two last decades, there has been much progress in developing the lattice Boltzmann method (LBM) as a powerful technique for computing fluid flow problems. The LBM is based on the kinetic theory in which it studies the particles interactions by using the particle mass distribution function. This mechanism is parallel in nature due to the locality of the dynamic of particles. From a computational point of view, it is suitable for massively parallel computing. Simplicity of programming and ease of considering microscopic interactions for modeling of additional physical phenomenon are the other advantages of the LBM. The standard LBM with the Bhatnagar–Gross–Krook (BGK) approximation is restricted to use uniform Cartesian grids with equal spacing. Inherent instability at high Reynolds number flows is also another limitation of this methodology. These two aspects in the standard LBM greatly limit its applications to solve engineering practical problems with complex geometries.

A drawback of using uniform grids is that it needs a high grid resolution in the high gradient regions, such as the thin boundary layer near a solid wall in a high Reynolds number flow. Thus, in order to solve high Reynolds number flows with a sufficient accuracy, a large number of uniform lattices should be used which requires larger computer resources [1]. The study of the treatment of curved or irregular boundaries and the control of mesh around complex geometries are also difficult with uniform Cartesian grids. In Navier–Stokes equation-based solvers, this issue is resolved by using non-uniform grids with mesh stretching techniques, often in a curvilinear coordinates system.

The difficulties and instabilities of the standard LBM from the use of uniform grids are closely related to the strong coupling of

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the time and space discretizations in the standard LBM. He and Luo [2] have shown that the discretization of the physical space does not necessarily require for coupling with the discretization of the momentum space. Based on this idea, alternative lattice Boltzmann (LB) methods are proposed based on non-uniform grids to overcome the limitations of the standard LBM to increase the numerical efficiency and accuracy of this methodology as a practical computational fluid dynamics tool. In this way, several extensive efforts have been made to provide improved versions of the LBM which can be classified into two different categories [3]: namely, interpolation types and differential types. In the interpolation strategy, the calculation is based on a coarse grid covering the whole integration domain. Then, a grid refinement technique is used to locally refine the grid in the critical regions where a high grid resolution is needed [3–5]. Another approach is to use a coupling of grids of varying resolution (non-uniform grids) by a local grid interface based on local interpolations [6–9].

In the differential types, the LBM is considered as a particular discretization of the discrete Boltzmann equation (DBE) which is transformed by using the Taylor series expansion in time and space [10]. Some efforts have been made in the literature to apply the traditional numerical methods developed in computational fluid dynamics for solving the DBE. Finite-difference (FD) [11–18], finite-volume (FV) [19–25], finite-element (FE) [26–28], spectral-element discontinuous Galerkin (SEDG) [29–31], and recently Chebyshev collocation spectral (CCS) [32] methods have been applied in order to improve computational accuracy and efficiency of the LBM. The differential type LBMs can overcome the difficulties encountered at high Reynolds numbers by using body-conform, curvilinear meshes with clustering of grid points in critical zones. Although the differential type LBMs provide such advantages, the locality of the collision of particles is sacrificed when applying these methods because of the stencil in discretizing the spatial derivative terms and the necessity of especial treatments near each boundary. Consequently, the compatibility with parallel computation, that is the main benefit of the standard LBM, is removed when using the differential type LBMs.

Among of existing numerical methods to solve the equations governed fluid flows, the spectral methods are indicated to be significantly more accurate with advantage of having good resolution characteristics. Handling curved boundaries with these methods is possible and can be made by applying a coordinate transformation. Recently, we have proposed and applied the Chebyshev collocation spectral lattice Boltzmann method (CCSLBM) to accurately solve incompressible flows in the Cartesian coordinates with the Chebyshev–Guass–Lobatto grid points [32]. The main objective of this paper is to implement the Chebyshev collocation spectral scheme to the LBM in the generalized curvilinear coordinates with non-uniform grids. As a result, the geometry of the curved boundaries can be more reasonably modeled and the solution of flow field is obtained with high-order accuracy. Note that, similar to other collocation spectral methods, especial treatments are needed for implementing boundary conditions. Herein, the spatial derivatives in the low-speed form of the LB equation in the generalized curvilinear coordinates are discretized by using the Chebyshev collocation spectral scheme and the temporal term is discretized by the fourth-order Runge–Kutta scheme to provide an accurate and efficient incompressible flow solver. A sensitivity study is also performed to examine the effects of the numerical parameters on the accuracy and performance of the solution. To assess the proposed method (CCSLBM) in terms of the accuracy and computational cost, a second-order finite-difference LBM (FDLBM) in the generalized curvilinear coordinates is also developed and systematic and quantitative comparisons between these two LBM solvers are made through solving steady and unsteady low-speed flow problems.

The paper is organized as follows: In Section 2, the low-speed form of the LB equation based on the pressure distribution function as the independent variable is presented and it is transformed into the generalized curvilinear coordinates in Section 3. The Chebyshev collocation spectral discretization and the second-order finite-difference discretization together with the Runge–Kutta time stepping scheme for the LB equation in the generalized curvilinear coordinates are given in Section 4. The boundary conditions required for the numerical scheme for both the macroscopic and microscopic variables are described in Section 5. Section 6 is devoted to present the 2D numerical results for different low-speed flow problems to examine the accuracy and performance of the solution of the Chebyshev collocation spectral LBM implemented in the generalized curvilinear coordinates. The extension of the solution methodology to 3D is given in Section 7 and then, the 3D numerical results are presented in Section 8. Finally, some conclusions are given in Section 9.

2. Governing equations

The main goal of this paper is to implement the Chebyshev collocation spectral lattice Boltzmann method (CCSLBM) in the generalized curvilinear coordinates. Here, the solution of the low-speed form of the LB equation is obtained using the Chebyshev collocation spectral method to enhance the computationally efficiency of the scheme implemented on curve boundaries with non-uniform grids. The Boltzmann equation governed the particle distribution function $f(t, \boldsymbol{e}, x)$ is utilized with the single relaxation-time and the Bhatnagar–Gross–Krook (BGK) approximation [33]:

$$\frac{\partial f}{\partial t} + \boldsymbol{e} \cdot \nabla f = -\frac{1}{\tau}(f - f^{eq}) \tag{1}$$

where $\boldsymbol{e}$ is the particle velocity,τ is the dimensionless collision relaxation-time, and f^{eq} is the equilibrium distribution function. With discretizing the velocity space $\boldsymbol{e}$, the LB equation for the particle distribution function f_k in the direction of microscopic velocity $\boldsymbol{e}_k$ may be written as

$$\frac{\partial f_k}{\partial t} + \mathrm{e}_k \cdot \nabla f_k = -\frac{1}{\tau}\left(f_k - f_k^{eq}\right), \quad k = 0, 1, ..., 8 \tag{2}$$

where the subscript k denotes the direction of the particle speed. In the D2Q9 discrete Boltzmann model, the microscopic velocities (see Fig. 1) are given as

$$\begin{aligned}\mathbf{e}_k &= \begin{bmatrix} e_{kx} \\ e_{ky} \end{bmatrix} = \begin{bmatrix} e_{0x} & e_{1x} & e_{2x} & e_{3x} & e_{4x} & e_{5x} & e_{6x} & e_{7x} & e_{8x} \\ e_{0y} & e_{1y} & e_{2y} & e_{3y} & e_{4y} & e_{5y} & e_{6y} & e_{7y} & e_{8y} \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 & 0 & -1 & 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 & -1 & 1 & 1 & -1 & -1 \end{bmatrix}\end{aligned} \tag{3}$$

In this formulation, the equilibrium distribution function is defined as [34]

$$f_k^{eq} = \alpha_k \left(p + p_0 \left(3(\mathrm{e}_k \cdot \mathrm{u}) + \frac{9}{2}(\mathrm{e}_k \cdot \mathrm{u})^2 - \frac{3}{2}(\mathrm{u} \,.\, \mathrm{u}) \right) \right) \tag{4}$$

with $\alpha_0 = 4/9$,$\alpha_1 = \alpha_3 = \alpha_5 = \alpha_7 = 1/9$, and $\alpha_2 = \alpha_4 = \alpha_6 = \alpha_8 = 1/36$ for the D2Q9 model, and $\boldsymbol{u} = (u, v)$ is the velocity vector.

The macroscopic pressure $p$and the macroscopic velocity vector u are obtained using the following relations:

$$p = \sum_{k=0}^{8} f_k, \quad p_0 \boldsymbol{u} = \sum_{k=0}^{8} \boldsymbol{e}_k f_k \tag{5}$$

where $p_0 = c_s^2 \rho_0$ and ρ_0 is the constant density of the fluid. The incompressible Navier–Stokes equations can be derived from the incompressible form of the LB model through the Chapman–Enskog expansion procedure [34]:

$$\frac{1}{c_s^2}\frac{\partial P}{\partial t} + \nabla\cdot\boldsymbol{u} = 0 \tag{6}$$

$$\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u}\cdot\nabla\boldsymbol{u} = -\nabla P + \upsilon\,\nabla^2\boldsymbol{u} \tag{7}$$

where $P = p/\rho_0$ is the normalized pressure and υ is the kinematic viscosity of the fluid. The relaxation-time τ in this model is defined by [16]

$$\tau = \frac{\upsilon}{c_s^2} \tag{8}$$

where $c_s = 1/\sqrt{3}$ is the speed of sound of the LB model.

3. Transformation to generalized curvilinear coordinates

Here, the LB Eq. (2) is transformed into the generalized curvilinear coordinates in which the physical and computational planes are denoted by (x, y) and (ξ, η), respectively:

$$\begin{aligned}\xi &= \xi(x, y)\\ \eta &= \eta(x, y)\end{aligned} \tag{9}$$

where the relationship between the physical and computational domains satisfies the following condition:

$$\begin{bmatrix}\xi_x & \xi_y\\ \eta_x & \eta_y\end{bmatrix} = \frac{1}{J}\begin{bmatrix}y_\eta & -x_\eta\\ -y_\xi & x_\xi\end{bmatrix} \tag{10}$$

and $\xi_x = \partial\xi/\partial x$, $\xi_y = \partial\xi/\partial y$, $\eta_x = \partial\eta/\partial x$ and $\eta_y = \partial\eta/\partial y$ are the metrics and the Jacobian of the transformation J is denoted by

$$J = x_\xi y_\eta - x_\eta y_\xi \tag{11}$$

The convection term in Eq. (2) by using the chain rule is rewritten as

$$\begin{aligned}\boldsymbol{e}_k\cdot\nabla f_k &= e_{kx}\frac{\partial f_k}{\partial x} + e_{ky}\frac{\partial f_k}{\partial y} = e_{kx}\left(\frac{\partial f_k}{\partial \xi}\xi_x + \frac{\partial f_k}{\partial \eta}\eta_x\right)\\ &\quad + e_{ky}\left(\frac{\partial f_k}{\partial \xi}\xi_y + \frac{\partial f_k}{\partial \eta}\eta_y\right)\\ &= \left(e_{kx}\xi_x + e_{ky}\xi_y\right)\frac{\partial f_k}{\partial \xi} + \left(e_{kx}\eta_x + e_{ky}\eta_y\right)\frac{\partial f_k}{\partial \eta}\\ &= \tilde{e}_{k\xi}\frac{\partial f_k}{\partial \xi} + \tilde{e}_{k\eta}\frac{\partial f_k}{\partial \eta}\end{aligned} \tag{12}$$

where $\tilde{\boldsymbol{e}}_k = (\tilde{e}_{k\xi}, \tilde{e}_{k\eta}) = (e_{kx}\xi_x + e_{ky}\xi_y, e_{kx}\eta_x + e_{ky}\eta_y)$ denotes the microscopic contravariant velocity vector in the computational plane. Now, Eq. (2) can be expressed in the computational plane as follows:

$$\frac{\partial f_k}{\partial t} + \left(\tilde{e}_{k\xi}\frac{\partial f_k}{\partial \xi} + \tilde{e}_{k\eta}\frac{\partial f_k}{\partial \eta}\right) = -\frac{1}{\tau}\left(f_k - f_k^{eq}\right), \quad k = 0, 1, \ldots, 8 \tag{13}$$

Note that the microscopic velocity vectors $\boldsymbol{e}_k$ are constant in the physical plane, however, the transformed microscopic velocity vectors in the computational plane, the contravariant velocity vectors $\tilde{\boldsymbol{e}}_k$, are not constant. Fig. 2 illustrates the vector $\boldsymbol{e}_5 = (e_{5x}, e_{5y}) = (1, 1)$ and $\tilde{\boldsymbol{e}}_5 = (\tilde{e}_{5\xi}, \tilde{e}_{5\eta})$ in the physical and computational planes, respectively, for the cylindrical Couette flow problem.

4. Discretization procedure

4.1. Chebyshev collocation spectral lattice Boltzmann method

In this study, the Chebyshev collocation spectral method is applied to solve the LB equation in the generalized curvilinear coordinates. Herein, the spatial derivatives in the LB equation in the computational plane are discretized by the Chebyshev collocation spectral method to obtain high accuracy solutions.

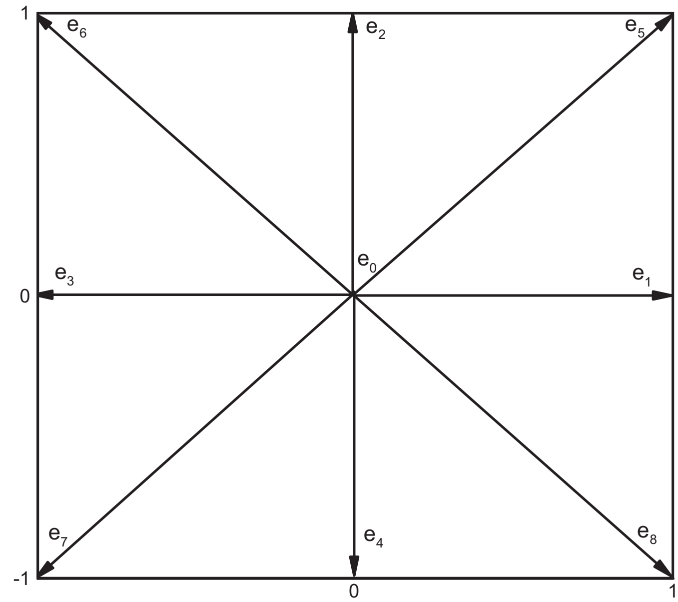


Fig. 1. The D2Q9 lattice and the microscopic velocities.

The spatial derivative of $u(\xi, t)$ at the collocation points ξ_k is evaluated using the following relation:

$$u^{(r)}(\xi_k) = \sum_{j=0}^{N} D_{kj}^{(r)} u(\xi_j), \quad r = 1, 2, \ldots \tag{14}$$

where $D_{kj}^{(r)}$ are the elements of the Chebyshev collocation derivative matrix, r is the order of differentiation and N is the polynomial degree that represents the number of grid points. The differentiation matrix $D_{kj}^{(1)}$ is given by

$$D_{kj}^{(1)} = \begin{cases}\frac{\lambda_j}{\lambda_k}\frac{1}{\xi_k - \xi_j} & if\ k \neq j\\ -\sum\limits_{i=0, i\neq k}^{N}\frac{\lambda_i}{\lambda_k}\frac{1}{\xi_k - \xi_i} & if\ k = j\end{cases} \tag{15}$$

where

$$\lambda_k^{-1} = \prod_{i=0, i\neq k}^{N}(\xi_k - \xi_i) \tag{16}$$

and the elements of the second derivative matrix are

$$D_{kj}{}^{(2)} = \begin{cases}2D_{kj}^{(1)}\left(D_{kk}^{(1)} - \frac{1}{\xi_k - \xi_j}\right) & if\ k \neq j\\ 2\left(D_{kk}^{(1)}\right)^2 + 2\sum\limits_{i=0, i\neq k}^{N} D_{ki}^{(1)}\frac{1}{\xi_k - \xi_i} & if\ k = j\end{cases} \tag{17}$$

The procedure given above for the calculation of the derivatives in one-dimension can be readily extended to two dimensions. If an unknown matrix U is defined as $U(\xi_i, \eta_j) = u_{ij}$, then its partial derivatives evaluated at the collocation points can be expressed in terms of the matrix–matrix products, where the differentiation with respect to ξ corresponds to multiplying the rows of D_ξ (the collocation derivative matrix in the ξ-direction) by the columns of U, and the differentiation with respect to η corresponds to multiplying the rows of U by the columns of D_η^T (the collocation derivative matrix transpose in the η-direction).

Now, by performing the space discretization using the spectral method, the LB Eq. (13) can be written in the following form:

$$\frac{\partial f_k}{\partial t} = R_k \tag{18}$$

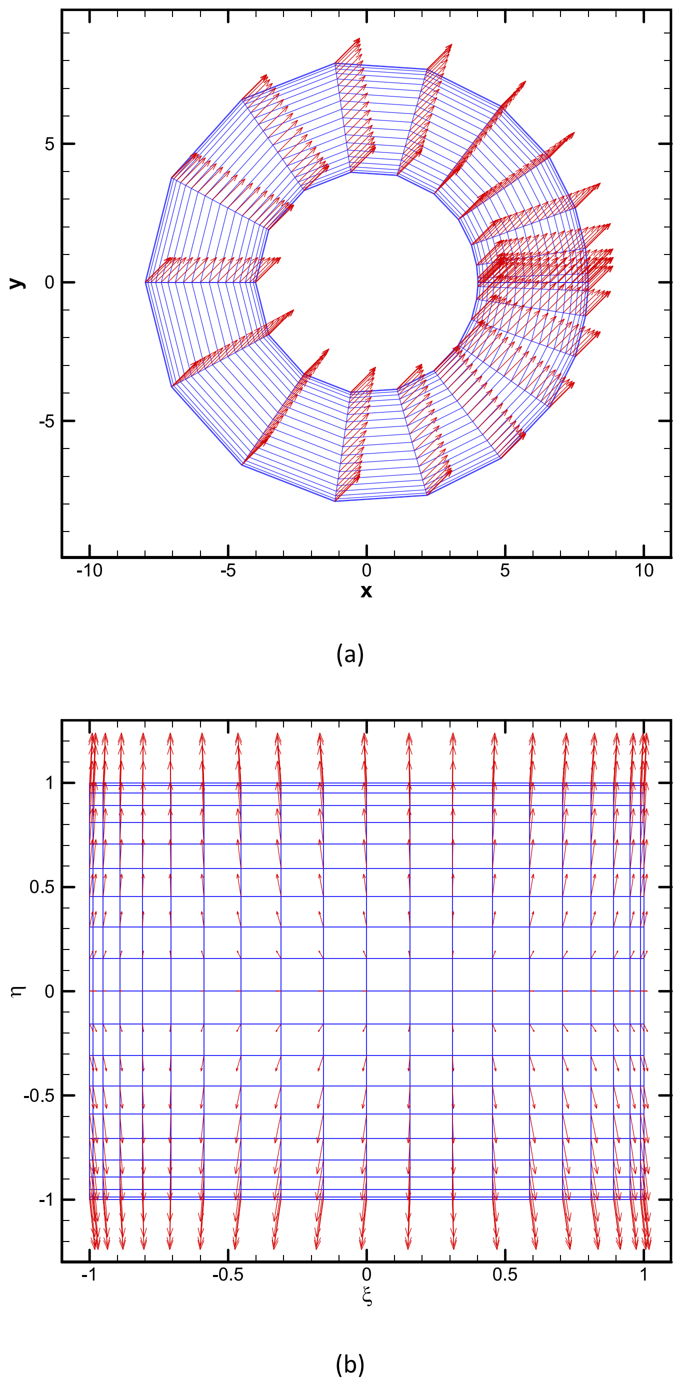


Fig. 2. (a) Microscopic velocity vector $e_5 = (e_{5x}, e_{5y}) = (1, 1)$ in the physical plane and (b) contravariant velocity vector $\tilde{e}_5 = (\tilde{e}_{5\xi}, \tilde{e}_{5\eta})$in the computational plane for cylindrical Couette flow problem.

where

$$R_{k_{m,n}} = -\left(\tilde{e}_{k\xi}\sum_{i=0}^{N_x} D_{\xi_{m,i}} f_{k_{i,n}} + e_{k\eta}\sum_{i=0}^{N_y} D^T_{\eta_{i,n}} f_{k_{m,i}}\right) - \frac{1}{\tau}\left(f_{k_{m,n}} - f^{eq}_{k_{m,n}}\right) \tag{19}$$

In Eq. (19), m and n indicate the grid number in the ξ- and η-direction, respectively, D_ξ is the derivative matrix in the ξ-direction and D^T_η is the transpose of the collocation derivative matrix in the η-direction. In addition, N_ξ and N_η are the polynomial degrees in each direction. Here, the metrices and the Jacobian of transformation for the problems simulated are calculated analytically.

The discretization of the temporal term in Eq. (18) is performed by an explicit multistage time-stepping method. Here, the solution is advanced in the time t by using the fourth-order Runge–Kutta scheme as follows:

$$\begin{aligned} f_k^0 &= f_k^t \\ f_k^1 &= f_k^0 + \frac{\Delta t}{4} R_k^0 \\ f_k^2 &= f_k^0 + \frac{\Delta t}{3} R_k^1 \\ f_k^3 &= f_k^0 + \frac{\Delta t}{2} R_k^2 \\ f_k^{t+\Delta t} &= f_k^0 + \Delta t\, R_k^3 \end{aligned} \tag{20}$$

This time integration scheme is appropriate for an accurate simulation of unsteady flows.

4.2. Finite difference lattice Boltzmann method

Here, a second-order finite-difference LBM (FDLBM) is also developed in the generalized curvilinear coordinates for the assessment of the accuracy and performance of the CCSLBM. In the FDLBM developed here, the spatial derivatives in the LB equation in the computational plane are discretized by a second-order central difference scheme and the temporal term is discretized with the fourth-order Runge–Kutta scheme. Now, by performing this space/time discretization, the right hand side of Eq. (18) can be written as

$$\begin{aligned} R_{k_{m,n}} = &-\left(\tilde{e}_{k\xi_{m,n}}\frac{f_{k_{m+1,n}} - f_{k_{m-1,n}}}{2\Delta\xi} + \tilde{e}_{k\eta_{m,n}}\frac{f_{k_{m,n+1}} - f_{k_{m-1,n-1}}}{2\Delta\eta}\right) \\ &- \frac{1}{\tau}\left(f_{k_{m,n}} - f^{eq}_{k_{m,n}}\right) - \overline{D_e} \end{aligned} \tag{21}$$

where $\overline{D_e}$ is the dissipation term defined as

$$\overline{D_e} = \frac{\varepsilon_e}{\Delta t}\left((\Delta\xi)^4\frac{\partial^4 f_k}{\partial\xi^4} + (\Delta\eta)^4\frac{\partial^4 f_k}{\partial\eta^4}\right) \tag{22}$$

and ε_e is the dissipation coefficient and the discretization of the fourth-order differential terms in Eq. (22) is performed with a fourth-order accuracy as follows:

$$\begin{aligned} \frac{\partial^4 f_k}{\partial\xi^4} &= \frac{f_{k_{m+2,n}} - 4f_{k_{m+1,n}} + 6f_{k_{m,n}} - 4f_{k_{m-1,n}} + f_{k_{m-2,n}}}{(\Delta\xi)^4} + O(\Delta\xi)^4 \\ \frac{\partial^4 f_k}{\partial\eta^4} &= \frac{f_{k_{m,n+2}} - 4f_{k_{m,n+1}} + 6f_{k_{m,n}} - 4f_{k_{m,n-1}} + f_{k_{m,n-2}}}{(\Delta\eta)^4} + O(\Delta\eta)^4 \end{aligned} \tag{23}$$

It should be noted that the FDLBM used here without applying the dissipation term does not provide a stable solution because of using a central difference scheme for the discretization of the spatial derivative term in the LB equation. Note that the dissipation term in the FDLBM should be used to damp the small spurious spatial scales generated by the boundary conditions and/or the flow non-linearities without affecting the large scale physical modes. A sensitivity study has been performed to examine the effect of the value of dissipation coefficient ε_e on the solution of the FDLBM. It was shown that very small values of ε_e may cause some oscillations in the solution whereas very large values of ε_e damp the solution and $\varepsilon_e = 0.001$ was selected as an appropriate value for the simulations performed by the FDLBM.

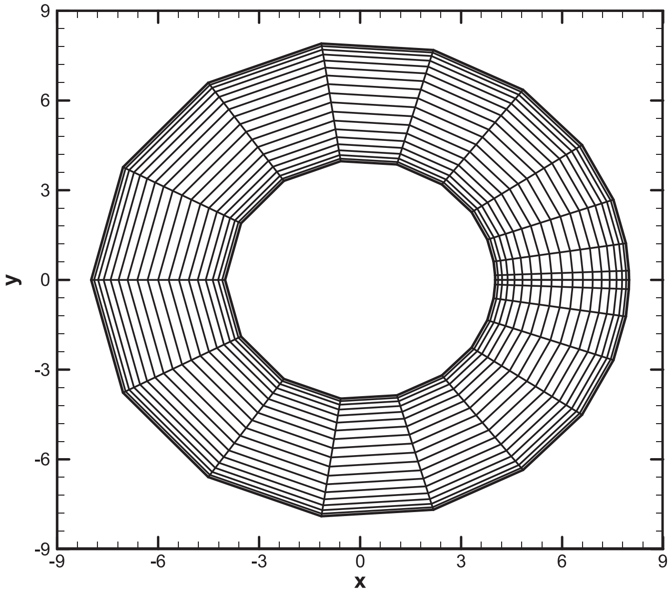


Fig. 3. A generated grid for the cylindrical Couette problem.

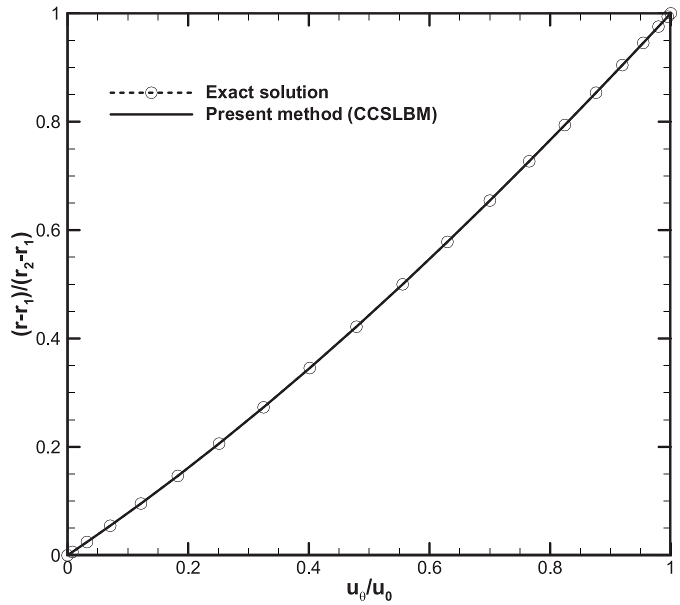


Fig. 5. Comparison of angular velocity profile for the steady cylindrical Couette flow.

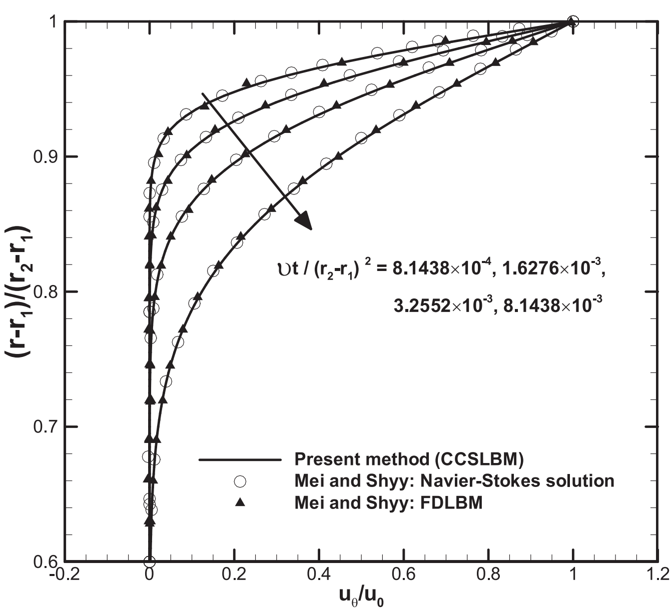


Fig. 4. Comparison of angular velocity profile for the unsteady cylindrical Couette flow at different non-dimensional times.

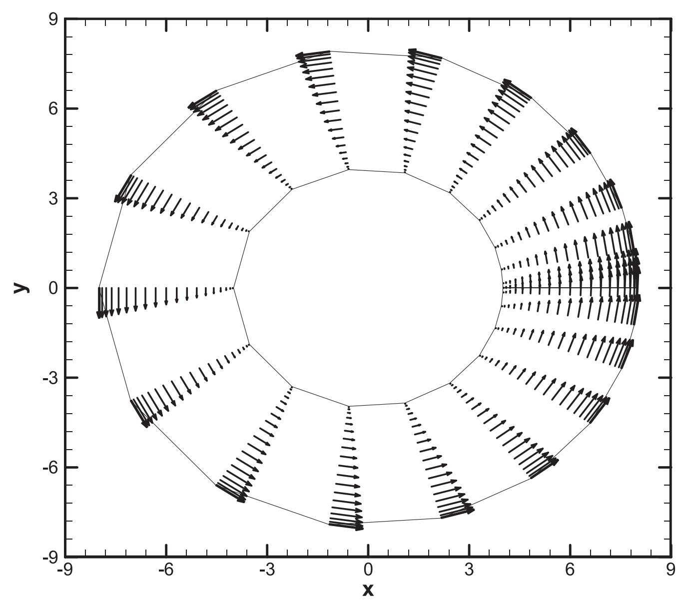


Fig. 6. Computed flow field shown by velocity vectors obtained by the CCSLBM for the steady cylindrical Couette flow.

5. Boundary conditions

In general, the solution of the spectral methods is quite sensitive to the boundary treatment and the algorithm may be unstable due to reflections of the numerical waves/errors from the boundaries into the flow field. Thus, the boundary conditions for the spectral methods should be implemented in a way to avoid the reflection of these numerical errors.

The Chebyshev collocation spectral method applied to solve the LB equation needs suitable boundary conditions for both the macroscopic and microscopic variables. In the LB equation, the distribution function f_k is not given directly at the boundaries and a special treatment should be utilized for determining its value based on the macroscopic variables on each boundary. Here, instead of any ad hoc boundary condition, the approach is to use the physical boundary conditions based on the solution of the governing equations implemented on each boundary to obtain accurate and stable solutions. On a wall boundary with the no-slip condition, the values of the macroscopic velocity components are known at each time step, $\boldsymbol{u} = (u, v) = (0, 0)$. For calculating the pressure on a wall boundary, one can obtain the following physical formula by employing the momentum equations at the wall:

$$\frac{\partial p}{\partial \eta} = \frac{1}{Re}\left[x_\eta\left(\gamma\frac{\partial^2 u}{\partial \eta^2}\right) + y_\eta\left(\gamma\frac{\partial^2 v}{\partial \eta^2}\right)\right], \quad \gamma = J^{-2}(x_\xi^2 + y_\xi^2) \quad (24)$$

where it is supposed that the subscript ξ is defined on the boundary wall. The same procedure can be used for calculating the pressure on the other boundaries. After updating the macroscopic variables, the equilibrium distribution function $f^{eq} = fun(u, v, p)$ at the boundaries is also updated by its formula, Eq. (4). Then, for calculating the distribution function f_k at a new time step, Eq. (2) is

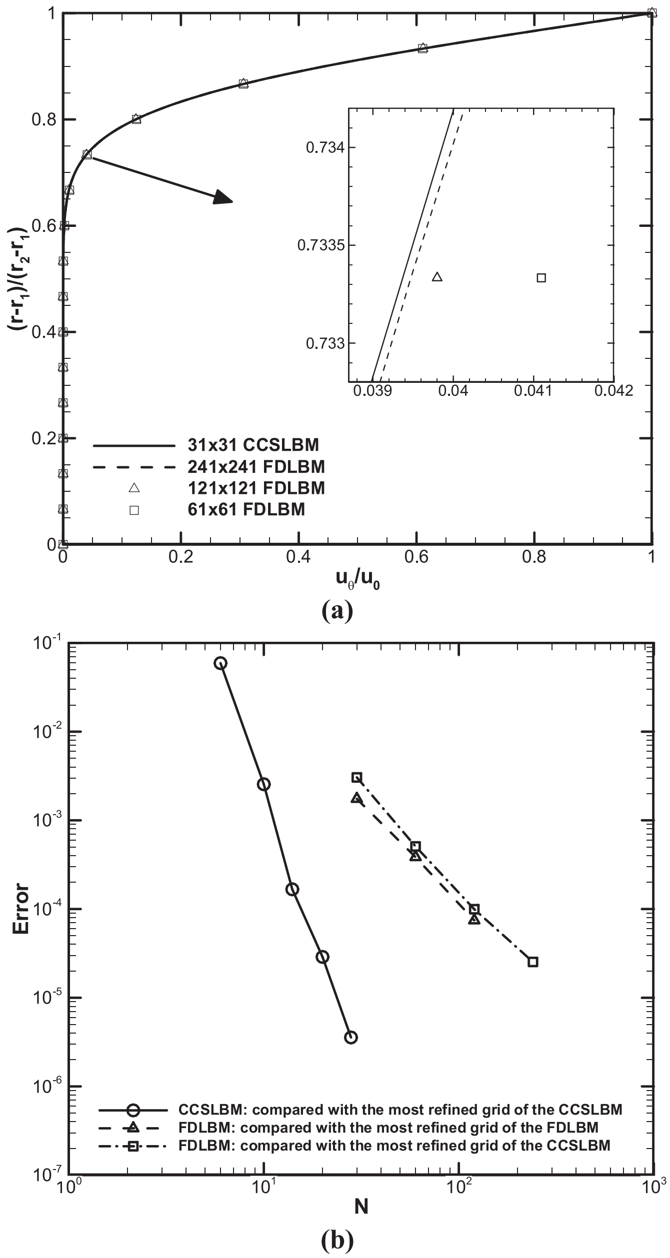


Fig. 7. Comparison of (a) angular velocity profile and (b) L_2-norm error of the solution obtained by the CCSLBM and the FDLBM for the unsteady cylindrical Couette flow at $t^* = 8.1438 \times 10^{-3}$.

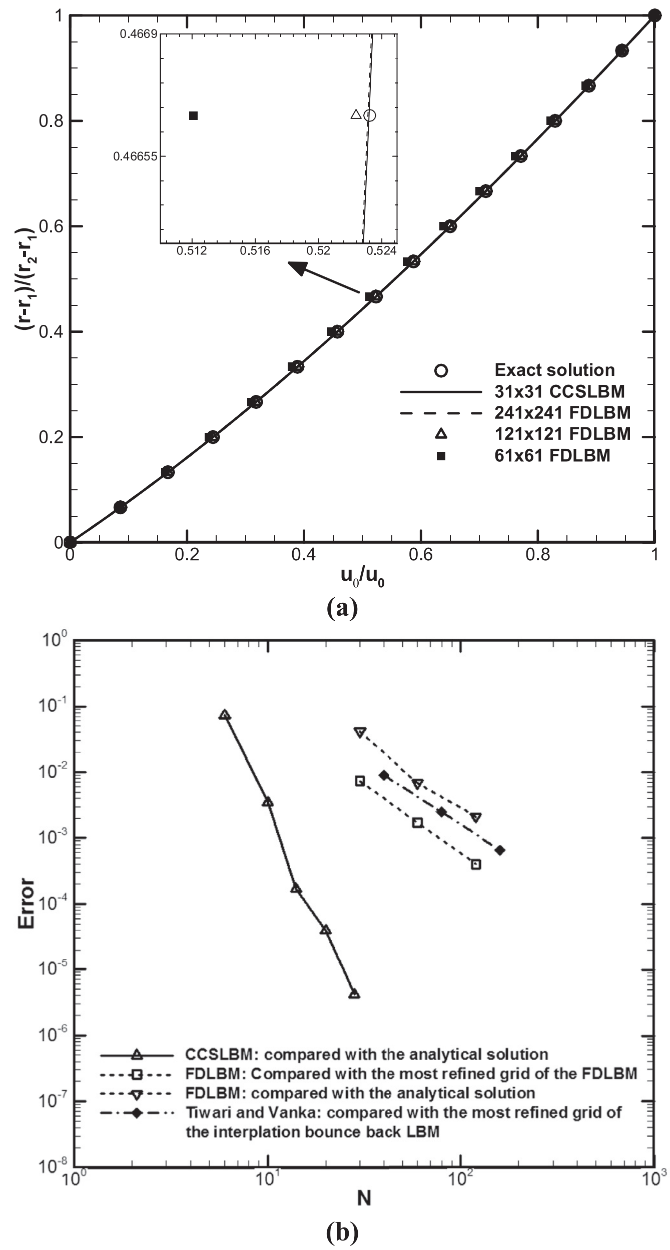


Fig. 8. Comparison of (a) angular velocity profile and (b) L_2-norm error of the solution obtained by the CCSLBM and the FDLBM for the steady cylindrical Couette flow.

solved at the boundaries with the same algorithm used for the interior points.

6. 2D numerical results

Herein, both steady and unsteady low-speed flows are simulated by applying the Chebyshev collocation spectral LBM (CCSLBM) in the generalized curvilinear coordinates. The Couette flow between concentric moving cylinders is simulated here to examine the accuracy of the solution algorithm for unsteady simulations. Results of the steady solution for this test case are also presented and compared with the analytical solution. For verifying the steady computations, two other benchmark test cases are considered: the regularized trapezoidal cavity flow and the flow in a gradual expansion duct. These two test cases are suitable problems to show that the CCSLBM is an effective solver to accurately compute steady flow problems with complex flow field structures and the solution methodology is robust and stable at different conditions. Results obtained for these problems by applying the CCSLBM are thoroughly compared and verified with the analytical solutions and those obtained in the literature by solving the Navier–Stokes equations. The results obtained by the Chebyshev collocation spectral LBM (CCSLBM) in the generalized curvilinear coordinates are also compared with those of the second-order finite-difference LBM (FDLBM) developed in terms of the accuracy and computational cost.

6.1. Unsteady and steady Couette flow between concentric moving cylinders

The low-speed flow between concentric moving cylinders with the radius of r_1 and r_2 is considered here to investigate the accuracy of the CCSLBM in solving unsteady and steady problems in the generalized curvilinear coordinate. At time $t < 0$, the fluid

Table 1
Comparison of error norm, CPU time and memory usage for the solution of the unsteady cylindrical Couette flow at $t^* = 8.1438 \times 10^{-3}$obtained by the CCSLBM with the FDLBM.

Grid	L_2	CPU time (h)	Memory usage (kilobytes)
CCSLBM			
7 × 7	5.9281E−02	0.003	592
11 × 11	2.5542E−03	0.007	648
15 × 15	1.6693E−04	0.016	724
21 × 21	2.8858E−05	0.055	884
29 × 29	3.5655E−06	0.115	1184
FDLBM			
31 × 31	1.7486E−03	0.045	1248
61 × 61	3.8515E−04	0.326	3320
121 × 121	7.4534E−05	1.365	10,448

Table 2
Comparison of error norm and CPU time for the solution of the steady cylindrical Couette flow obtained by the CCSLBM with the FDLBM.

Grid	L_2	CPU time (h)
CCSLBM		
7 × 7	7.2514E−02	0.056
11 × 11	3.4813E−03	0.160
15 × 15	1.6851E−04	0.822
21 × 21	3.9086E−05	1.764
29 × 29	4.1246E−06	4.485
FDLBM		
31 × 31	4.15E−02	3.657
61 × 61	6.86E−03	39.500
121 × 121	2.11E−03	175.657

is at rest and for $t \geq 0$, the upper cylinder suddenly moves with the angular velocity u_θ. Here, $r_1 = 4$, $r_2 = 8$, $u_0 = u_\theta = 0.005$, and $Re = u_0(r_2 - r_1)/\upsilon = 10$ are used to simulate this problem. The generated grid for $N_x = N_y = 20$ is illustrated in Fig. 3 and in both the radial and angular directions, the Chebyshev–Gauss–Lobatto points are used. Fig. 4 shows the unsteady solution of the Couette flow between concentric moving cylinders for the angular velocity profile at different non-dimensional times, $t^* = \upsilon t/H^2 = 8.1438 \times 10^{-4}$, 1.6276×10^{-3}, 3.2552×10^{-3}, and $8.1438 \times 10^{-3}$and good agreement between the solution of the CCSLBM and those obtained by Mei and Shyy [12] using the Navier–Stokes equations and the finite-difference LBM is obtained.

For the steady cylindrical Couette flow, the exact solution for the angular velocity is as follows:

$$\frac{u_\theta}{u_0} = \left(\frac{r_2}{r_1} - \frac{r_1}{r_2}\right)^{-1}\left(\frac{r}{r_1} - \frac{r_1}{r}\right) \tag{25}$$

Fig. 5 shows the steady angular velocity profile at the collocation points that indicates the CCSLBM has good agreement with the analytical solution. The computed flow field shown by the velocity vectors is represented in Fig. 6.

Now, the accuracy and performance of the proposed solution methodology based on the CCSLBM are examined by comparison with the FDLBM developed for the Couette flow between concentric moving cylinders. Fig. 7(a) shows a comparison of the unsteady-state angular velocity profile for this problem with $Re = 10$ at $\upsilon t/H^2 = 8.1438 \times 10^{-3}$ obtained by the CCSLBM and the FDLBM for different grid sizes and the corresponding L_2-norm errors for these two LBM solvers are shown in Fig. 7(b). Note that this problem for the unsteady case has not the exact solution for the calculation of the solution error. Here, the solution of the CCSLBM with the most refined grid is considered as the benchmark to calculate the error of the unsteady solution of the CCSLBM and the FDLBM for each grid. The solution of the FDLBM for each grid

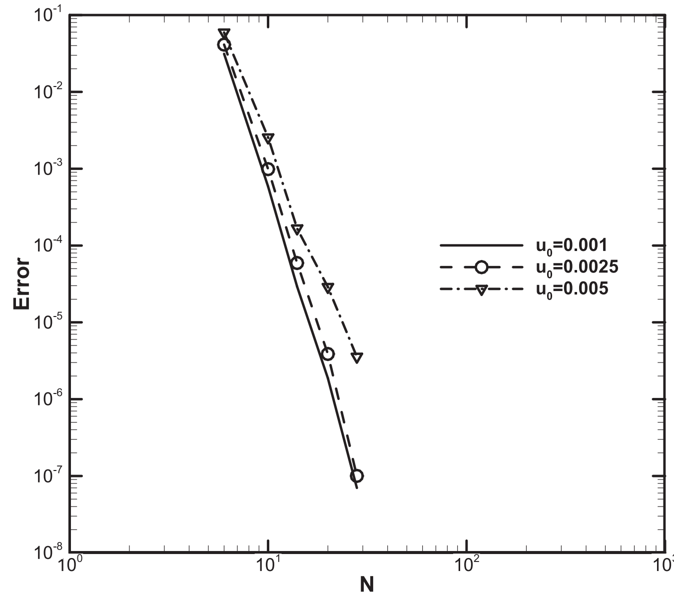


Fig. 9. Effect of characteristic velocity magnitude on L_2 norm error of the solution obtained by the CCSLBM for the unsteady cylindrical Couette flow at $t^* = 8.1438 \times 10^{-3}$.

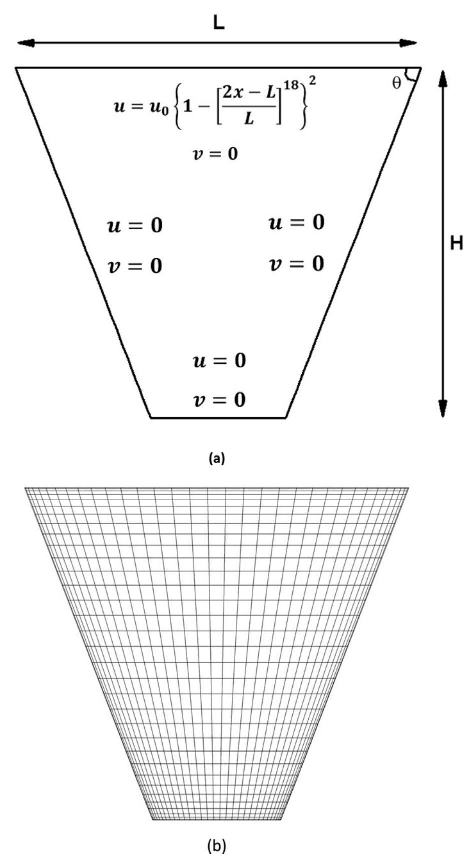


Fig. 10. (a) Schematic and (b) a generated grid for the 2D regularized trapezoidal cavity problem.

is also compared with the most refined grid of the FDLBM to calculate its error. Small differences are observed between the errors of the FDLBM when the solution of the most refined grid of the CCSLBM and the FDLBM is considered as the benchmark. The same study is performed for the steady case and the corresponding results are given in Fig. 8(a) and (b). Note that the error of the CC-

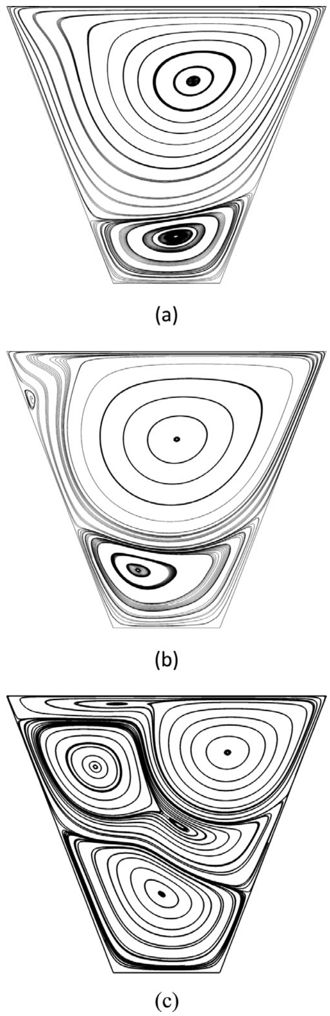


Fig. 11. Computed flow field shown by streamlines obtained by the CCSLBM for the first 2D regularized trapezoidal cavity flow at (a) $Re = 100$, (b) $Re = 500$ and (c) $Re = 1000$.

SLBM and the FDLBM for the steady case is calculated based on the angular velocity profile compared to the exact solution given by Eq. (25). The error of the FDLBM for each grid is also calculated based on the solution of the most refined grid of the FDLBM. Here, the accuracy of the solution of the CCSLBM and the FDLBM is also examined with that of the interpolation bounce back LBM performed by Tiwari and Vanka [35]. They have solved the steady cylindrical Couette flow using 41×41, 81×81 and 161×161 grid points by applying the interpolation bounce back LBM and the L_2 norm error of the angular velocity profile compared with the most refined grid 321×321 has been reported in their study. In the interpolation bounce back LBM [36], a Cartesian grid covers the geometry and thus some grid points are not placed in the solution domain and such solution procedure has a larger error than the FDLBM and the CCSLBM implemented in the curvilinear coordinates (see Fig. 8(b)). The figure also indicates that the CCSLBM applied in the curvilinear coordinates has the exponential convergence providing a more accurate solution than the FDLBM and the interpolation bounce back LBM by considering the same number of grid points. Unlike the FDLBM, the CCSLBM applied does not need any numerical dissipation or filtering for the solution to be stable and thus highly accurate solutions obtained by the CCSLBM can be used to assess the accuracy and performance of the other LBM-based solvers. Tables 1 and 2 compare the computational efficiency of the CCSLBM and the FDLBM in terms of the CPU time and the

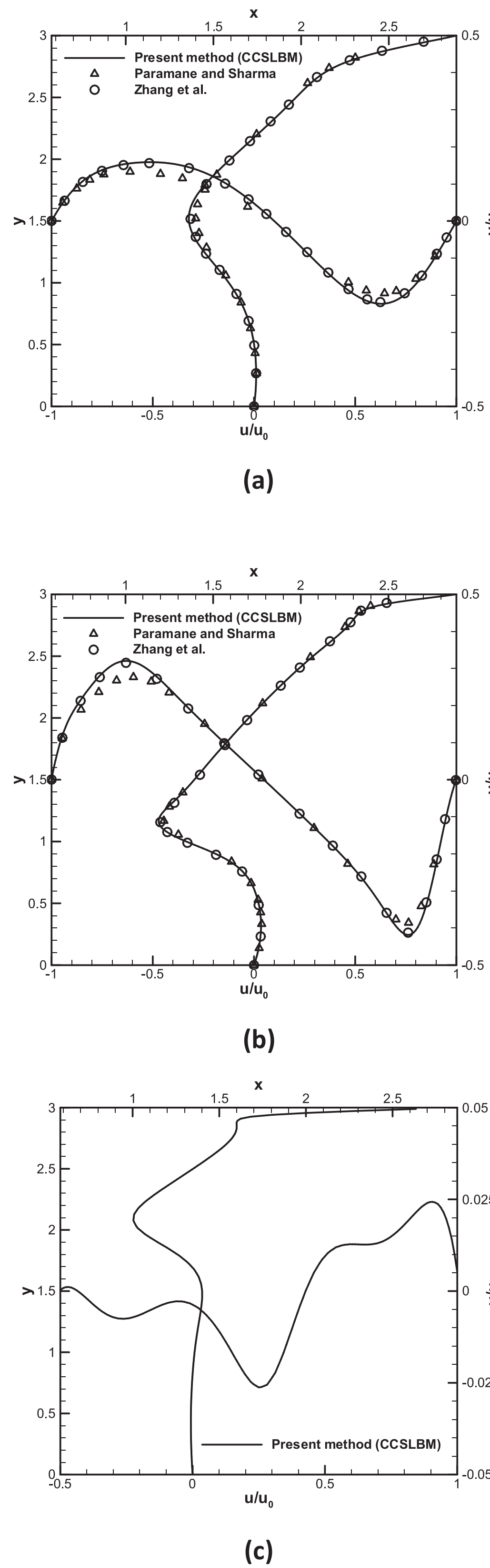


Fig. 12. Comparison of velocity profiles at midplanes of the first 2D regularized trapezoidal cavity flow at (a) $Re = 100$, (b) $Re = 500$ and (c) $Re = 1000$.

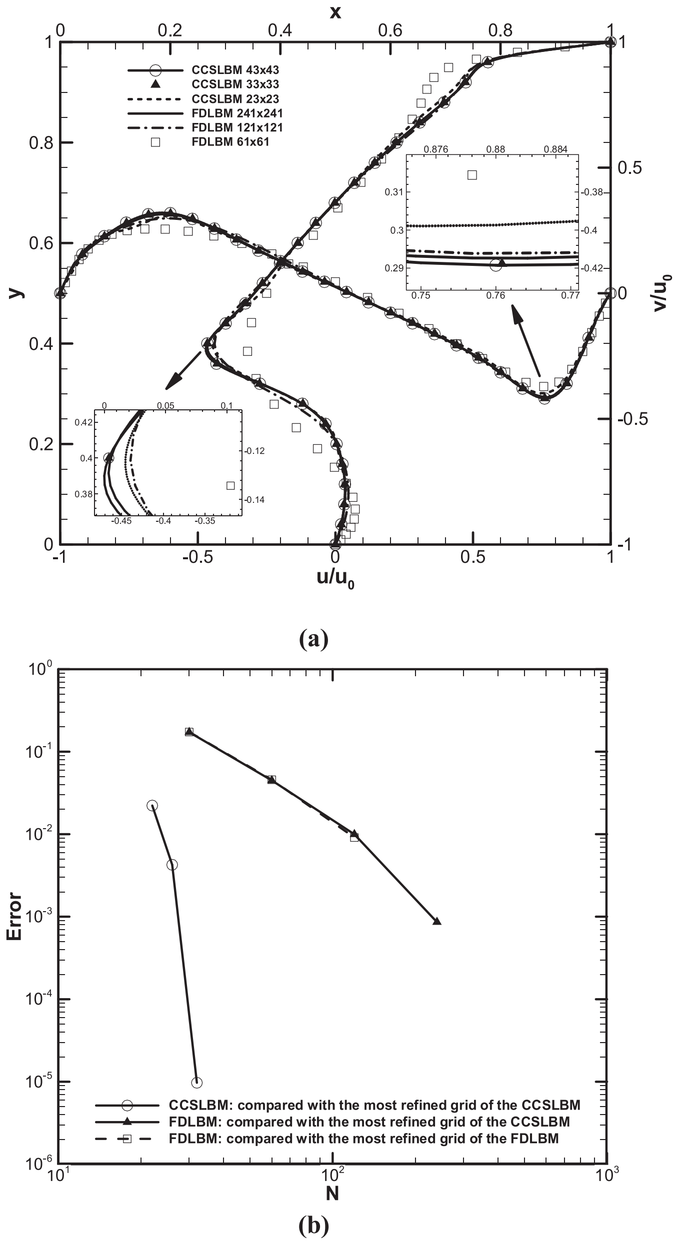


Fig. 13. Comparison of (a) velocity profiles at midplanes and (b) L_∞-norm error of the solution obtained by the CCSLBM and the FDLBM for the first 2D regularized trapezoidal cavity flow at $Re = 500$.

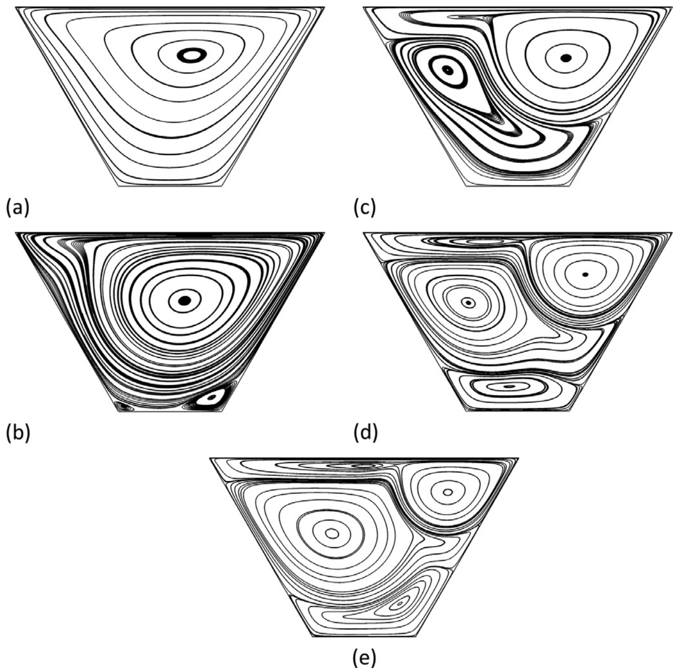


Fig. 14. Computed flow field shown by streamlines obtained by the CCSLBM for the second 2D regularized trapezoidal cavity flow at (a) $Re = 100$, (b) $Re = 1000$, (c) $Re = 2000$, (d) $Re = 3200$ and (e) $Re = 5000$.

Table 3
Comparison of error norm, CPU time and memory usage for the solution of the first 2D regularized trapezoidal cavity at $Re = 500$ obtained by the CCSLBM with the FDLBM.

CCSLBM : compared with most refined grid			
Grid	L_∞	CPU time (h)	Memory usage (kilobytes)
22	2.226E–02	0.064925	676
26	4.259E–03	0.4437	944
32	9.704E–06	1.183563	1376
FD : compared with most refined grid			
Grid	L_∞	CPU time (h)	Memory usage (kilobytes)
30	1.733E–01	0.799375	1296
60	4.572E–02	2.06474	3424
120	9.122E–03	14.14873	10,708
FD : compared with CCSLBM			
Grid	L_∞	CPU time (h)	Memory usage (kilobytes)
30	1.724E–01	0.799375	1296
60	4.486E–02	2.06474	3424
120	9.982E–03	14.14873	10,708
240	8.600E–04	56.88913	33,768

memory usage for the unsteady and steady cases, respectively. The error decays at faster rate by applying the CCSLBM and a lower number of grid points is needed to achieve a specified accuracy compared to the FDLBM and consequently the computational cost required for the CCSLBM is considerably decreased. It is also found that when a high accuracy solution is needed, the performance of the CCSLBM is more highlighted.

It should be noted that better computational efficiency of the CCSLBM over the FDLBM examined here is based on the computations performed on a single processor. Note that the spectral methods suffer from the requirement of the global data communications when they are implemented in the parallel codes and their computational efficiency in terms of the physical time may be decreased in big computations. The implementation of the multidomain technique to the CCSLBM can reduce the data dependencies and a better scalability of the solution method can be achieved in parallel and high performance computing. Note also that by applying the multidomain technique to the CCSLBM, the efficiency and flexibility of the CCSLBM in simulating the flow over more practical problems are improved.

Here, the sensitivity of the solution to the Mach number (or the characteristic velocity magnitude) is studied. For this purpose, three different characteristic velocity magnitudes, $u_0 =$ 0.005, 0.0025, and 0.001, are used to investigate the effect of the value of this parameter on the accuracy of the solution of the CCSLBM. Here, the L_2 norm error based on the velocity profiles at $t^* = 8.143 \times 10^{-3}$ is calculated, as shown in Fig. 9. The study indicates that smaller characteristic velocity magnitudes reduce the compressibility effects and thus more accurate results can be achieved.

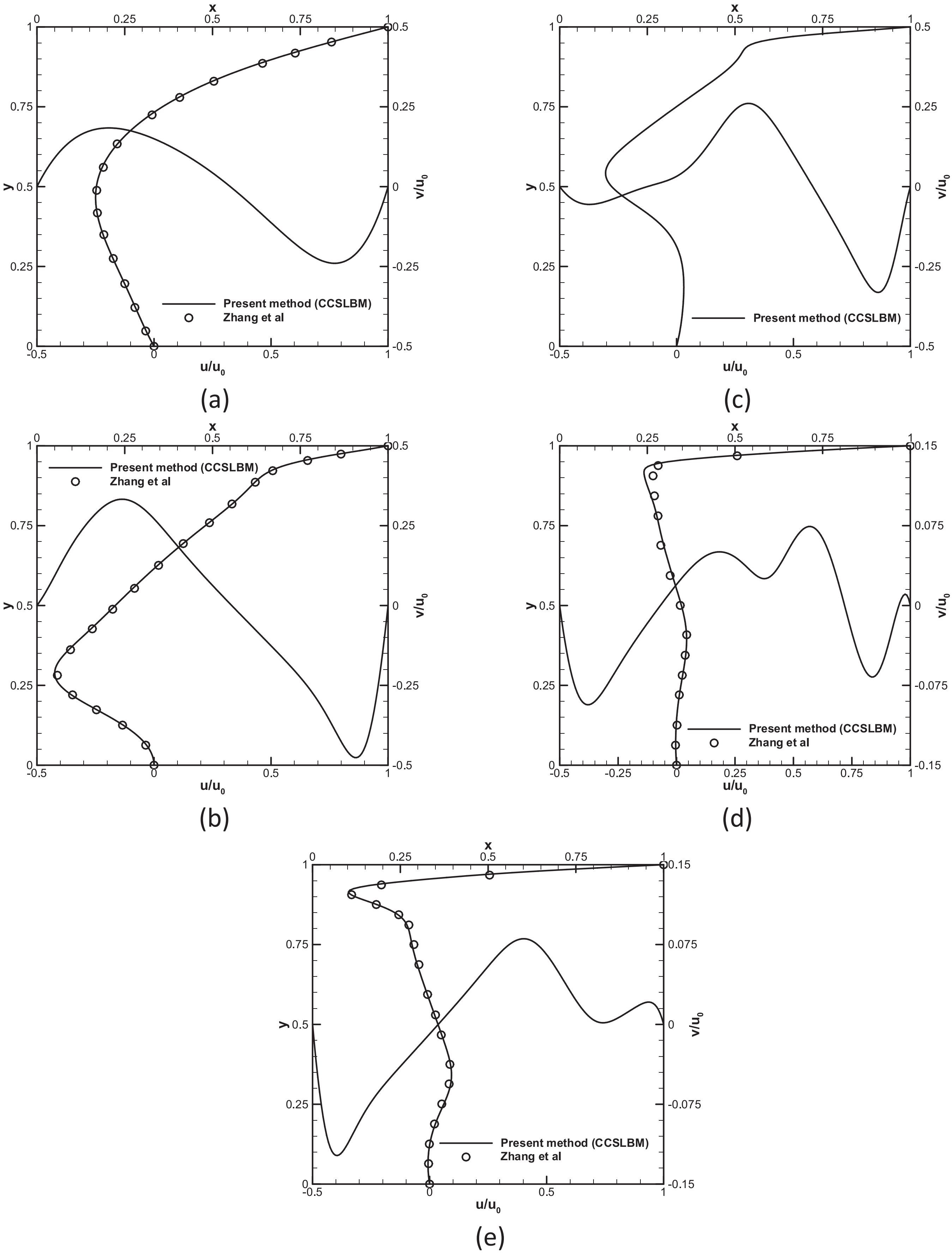


Fig. 15. Comparison of velocity profiles at midplanes of the second 2D regularized trapezoidal cavity flow at (a) Re = 100, (b) Re = 1000, (c) Re = 2000, (d) Re = 3200 and (e) Re = 5000.

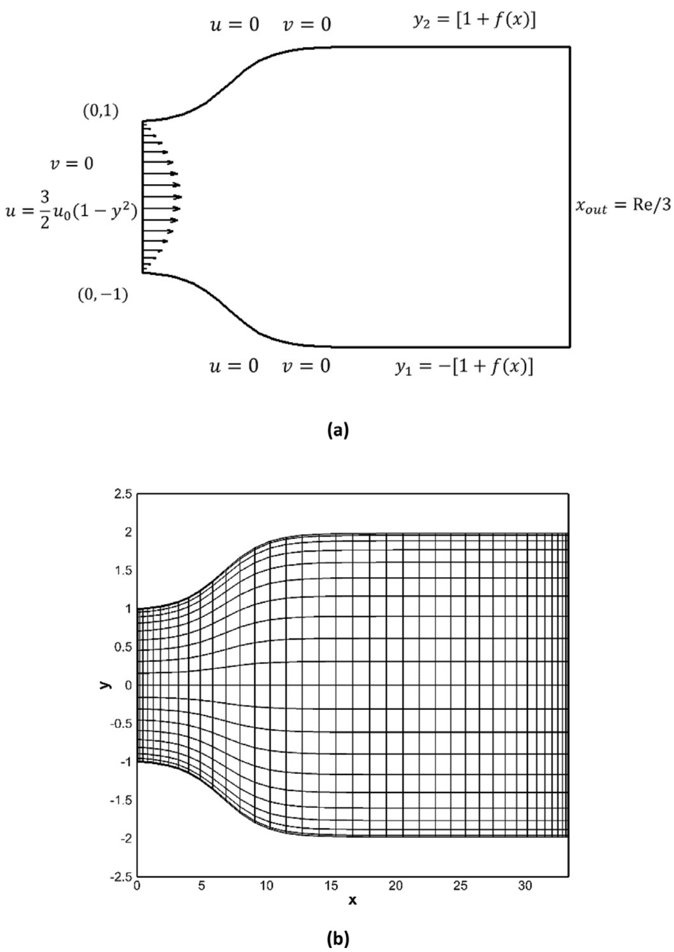


Fig. 16. (a) Schematic and (b) generated grid for the 2D flow in the gradual expansion duct problem.

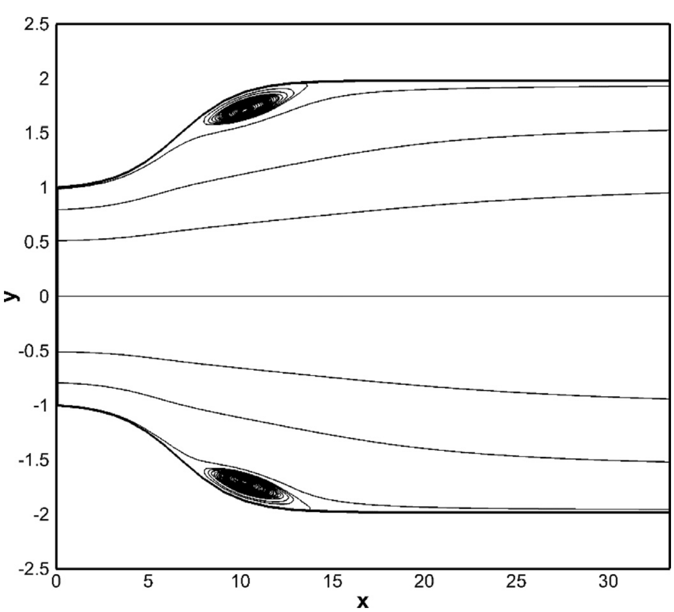


Fig. 17. Computed flow field shown by streamlines for the 2D flow in the gradual expansion duct problem at $Re = 100$.

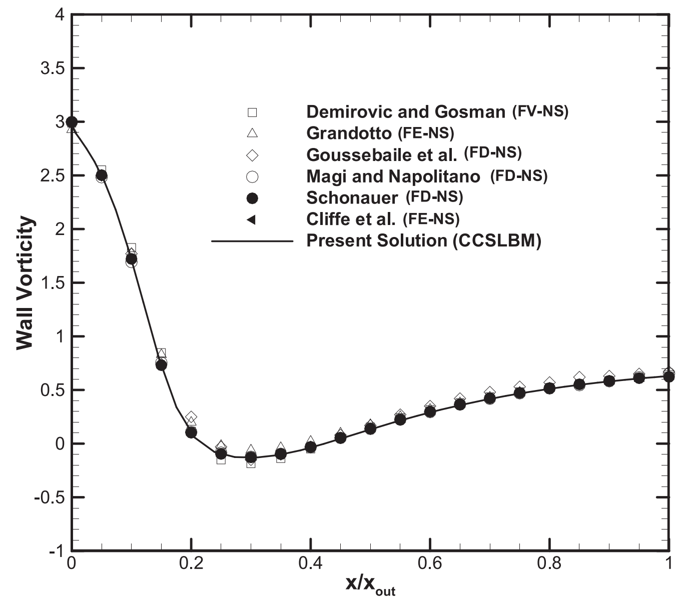


Fig. 18. Comparison of wall vorticity for the 2D flow in the gradual expansion duct problem at $Re = 100$.

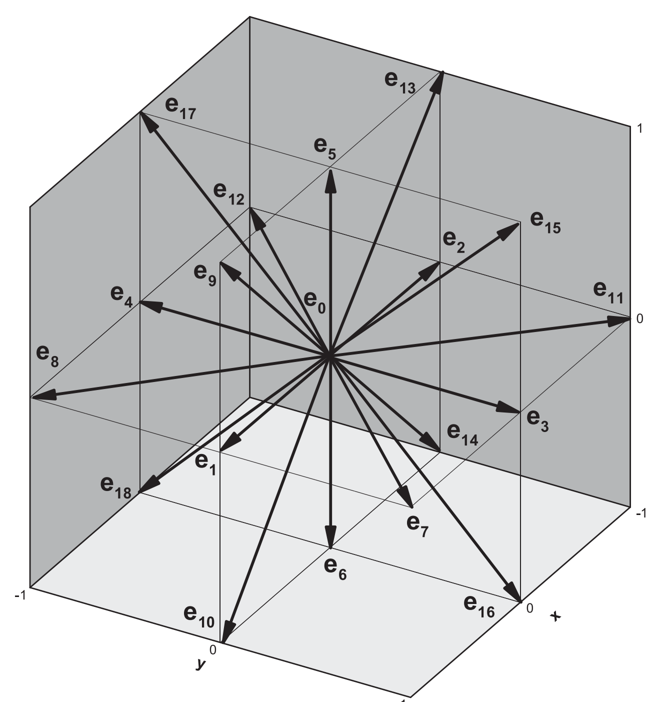
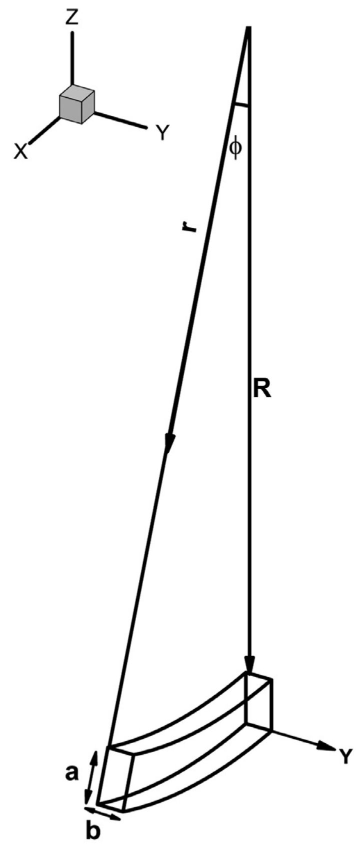


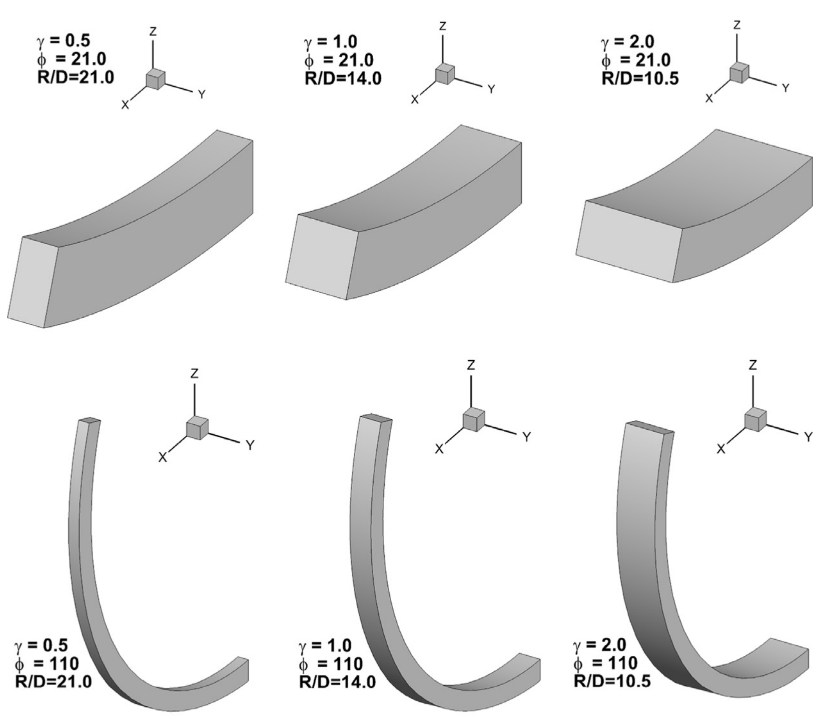
Fig. 19. The D3Q19 lattice and the microscopic velocities.

6.2. Steady flow in 2D regularized trapezoidal cavity

In order to show the high accuracy of the Chebyshev collocation spectral LBM proposed, two different 2D regularized trapezoidal cavity flow problems are simulated. In these problems, the singularity at the upper corners is resolved using a polynomial horizontal velocity distribution instead of a constant velocity. The geometry of the 2D regularized trapezoidal cavity and the grid used are shown in Fig. 10, The first 2D trapezoidal cavity has $L = 2\sqrt{3}$, $H = 3$ and $\theta = 90 - \tan^{-1}(2/(3\sqrt{3}))$ corresponding to the length, height and skewed angle between the top and right walls, respectively, and the second 2D trapezoidal cavity has $L = 1$, $H = \sqrt{3}/3$ and $\theta =$



(a)



(b)

Fig. 20. Geometry description of the 3D curved duct (a) in terms of the toroidal coordinate system and (b) different investigated geometries.

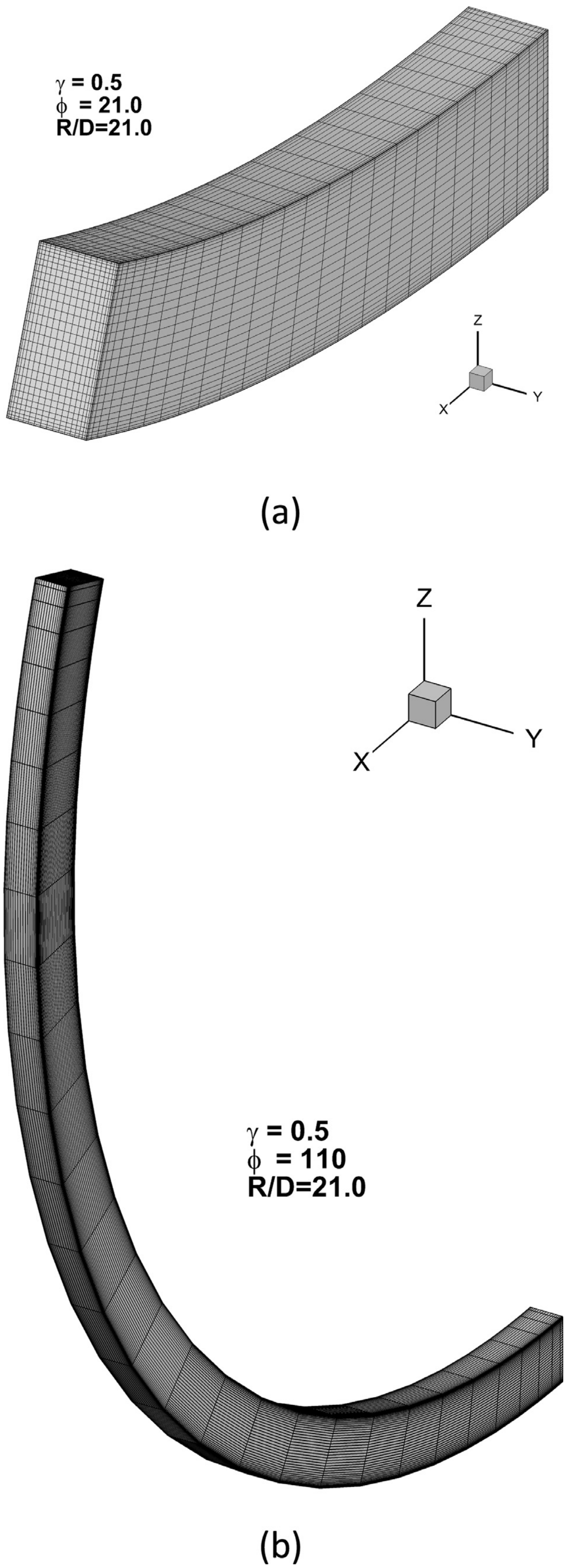


Fig. 21. Generated grid for the 3D curved duct problem for Case 1 for (a) $\phi = 21^{\circ}$ and (b) $\phi = 110^{\circ}$.

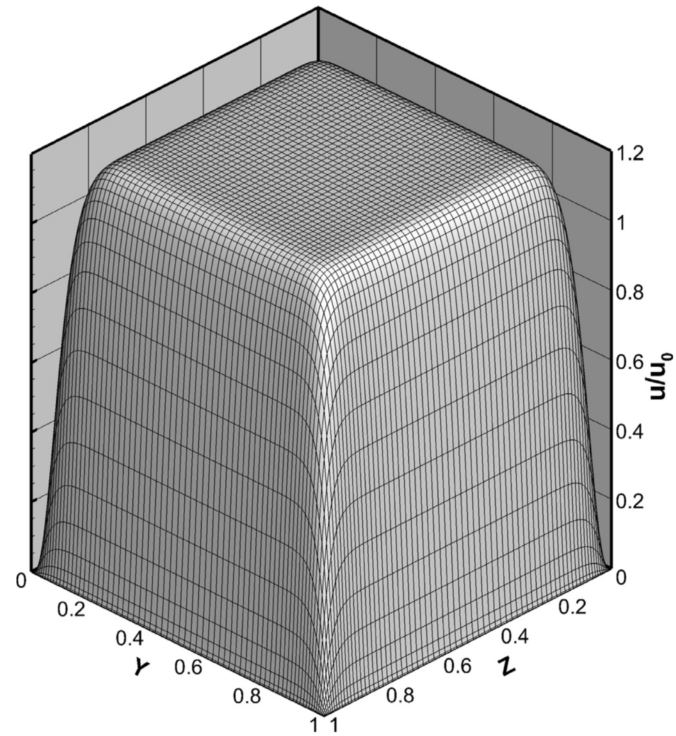


Fig. 22. Inlet velocity distribution for the 3D curved duct problem.

60. The boundary conditions for these problems are $u = v = 0$ for the left, right, and bottom walls and $u = u_0[1 - (\frac{2x-L}{L})^{18}]^2$, $v = 0$ for the top wall. The Reynolds number is defined as $Re = u_0 H/(3\upsilon)$ and $Re = u_0 L/\upsilon$ for the first and second trapezoidal cavity problems, respectively, and $u_0 = 0.1$ is set for all the simulations performed. Here, the time step $\Delta t = \tau$ is adopted to obtain a stable solution at high Reynolds number flows.

Before we proceed any further, the CCSLBM is validated by comparing the present results for the first regularized trapezoidal cavity with those of the standard LBM and the QUICK finite volume scheme reported in the previous studies [36,37]. The steady-state streamlines of the first regularized trapezoidal cavity problem at Re = 100, 500 and 1000 are shown in Fig. 11. For Re = 100 and 500 a mesh of 33 × 33 and for Re = 1000 a mesh of 81 × 81 are used to accurately compute the flow field for the first 2D regularized trapezoidal cavity test case. It is observed that with increasing Re from 100 to 500, the center of the primary vortex is moved toward the center of the cavity and a secondary vortex is created near the left wall. In addition, the center of the vortex near the lower wall is also shifted to the left wall. It is indicated that the number of vortices increase from three to five by increasing the Reynolds number from 500 to 1000; three large vortices are observed in the bottom, middle and top parts of the cavity and two small vortices in the middle and top parts of the cavity. It is illustrated that the complexity of the flow structure at $Re = 1000$ is accurately resolved by the CCSLBM. In Fig. 12, the steady-state velocity profiles along the vertical and horizontal lines through the cavity center at Re = 100, 500 and 1000 are presented and the results exhibit excellent agreement with the available numerical results [36,37].

To further assess the accuracy and performance of the CCSLBM in the generalized curvilinear coordinate, the first regularized trapezoidal cavity problem at Re = 500 is also simulated by the FDLBM and the results of these two LBM solvers are compared with each other. Fig. 13(a) compares the steady-state velocity profiles at the midplanes of the first regularized trapezoidal cavity at

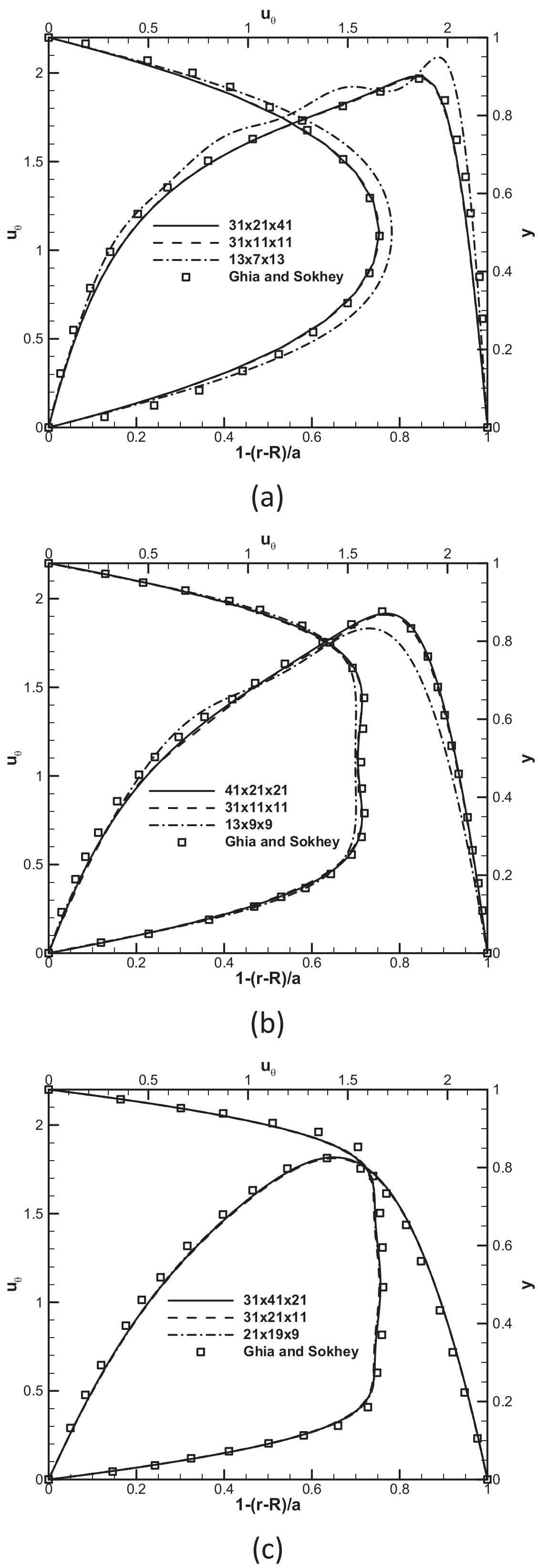


Fig. 23. Comparison of outlet velocity profiles at midplanes of the 3D curved duct problem for $\phi = 110^o$ with (a) $\gamma = 0.5$, (b) $\gamma = 1.0$, and (c) $\gamma = 2.0$.

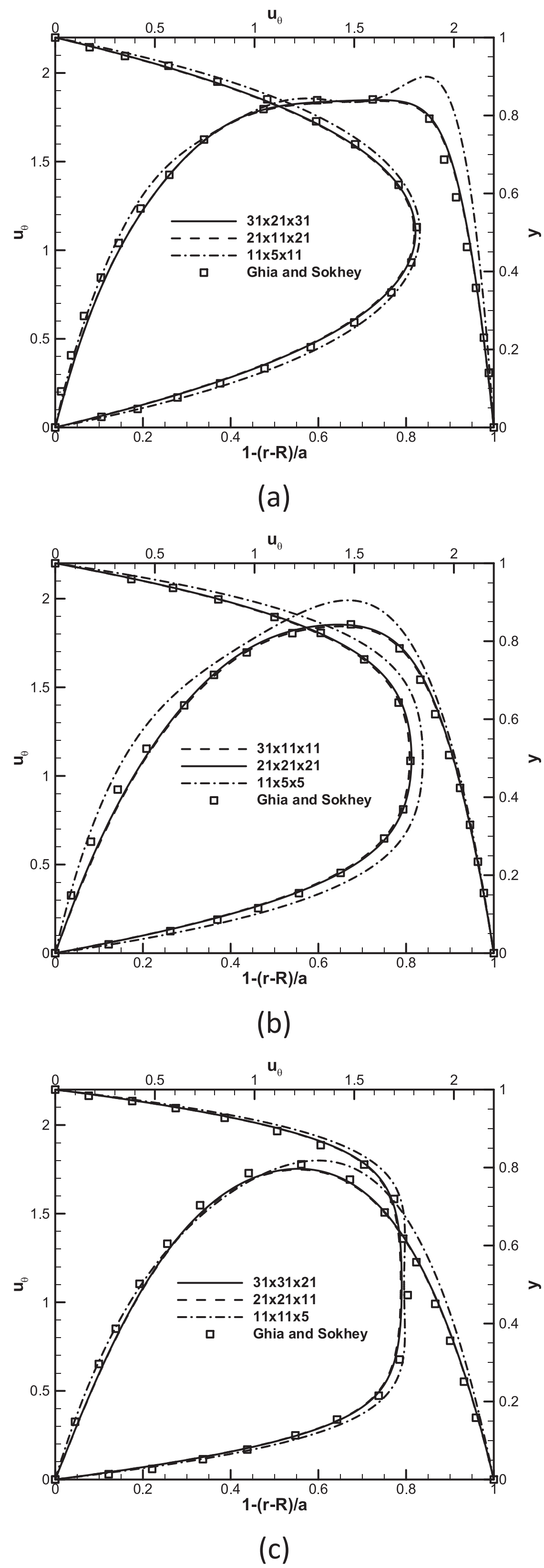


Fig. 24. Comparison of outlet velocity profiles at midplanes of the 3D curved duct problem for $\phi = 21^o$ with (a) $\gamma = 0.5$, (b) $\gamma = 1.0$, and (c) $\gamma = 2.0$.

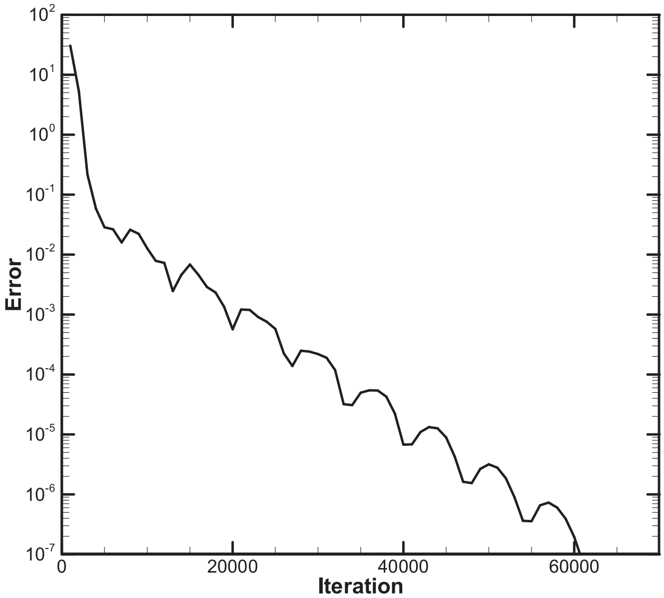


Fig. 25. Convergence history of solution for the 3D curved duct problem for $\phi = 21^{o}$ with $\gamma = 2.0$.

Re = 500 obtained by the CCSLBM and the FDLBM for different grid sizes and the corresponding solution errors for these two LBM solvers are given in Fig. 13(b). Note that for this test case, the exact solution is not available for the calculation of the solution error. Here, the error for both the CCSLBM and the FDLBM is calculated based on the *u*-velocity value at the center of the trapezoidal cavity compared to that of the most refined grid obtained by the CCSLBM, as shown in Fig. 13(b). In this figure, it is indicated that the solution error of the FDLBM is nearly the same when the solution of the most refined grid of the CCSLBM and the FDLBM is considered as the benchmark. It is observed that by considering the same number of grid points, the CCSLBM gives more accurate results than the FDLBM and the corresponding solution convergence rate is faster than that of the FDLBM. The computational efficiency of the proposed solution methodology based on the CCSLBM compared to the FDLBM can be concluded in Table 3. It is demonstrated that the CCSLBM considerably reduces the CPU time and memory usage required in comparison with the FDLBM and the performance of the CCSLBM is more highlighted when a high accuracy solution is desirable.

To accurately compute the flow field in the second 2D regularized trapezoidal cavity, for the Reynolds numbers 100 and 1000 a mesh of 33 × 33, for *Re* = 2000 a mesh of 41 × 41, for *Re* = 3200 a mesh of 81 × 81 and for *Re* = 5000 a mesh of 101 × 101 are used. This test case is simulated here to examine the accuracy and robustness of the CCSLBM in computing high Reynolds number flows. The computed flow field shown by the streamlines for the second 2D regularized trapezoidal cavity are shown in Fig. 14 for different *Re*. It is observed that the number of vortices in the cavity increase with increasing *Re*. For *Re* = 100, only one vortex is observed in the cavity. When *Re* is increased to 1000, two corner vortices are observed at the bottom of the cavity and the size of the right corner vortex is larger than the left one. For *Re* = 2000, the two corner vortices joint together and form a bigger vortex placed near the left wall. When *Re* is further increased to 3200 and 5000, the flow structure in this 2D regularized trapezoidal cavity becomes more complex; three large vortices are observed in the cavity and they are distributed in the top, middle and bottom parts of the cavity. In addition, one vortex that is smaller than the other three vortices is placed near the upper wall. By increasing *Re* from 3200 to 5000, the center of the bottom vortex shifts to the right and the size of the middle vortex is increased whereas the size of the top vortex is decreased. Fig. 15 demonstrates the computed flow field shown by the velocity profiles obtained by the CCSLBM in comparison with those of Zhang et al. [37] based on the interpolation bounce back LBM which show good agreement.

6.3. *Steady flow in a 2D gradual expansion duct*

In the occasion of the workshop-session of the fifth international association for hydraulics research (IAHR) meeting, it was decided to devote attention to flows in ducts of complex geometries, in order to permit the specialists to test the capabilities of different numerical schemes in dealing with this kind of flow simulations. In [38, 39], then it was recommended a typical test case well-indicated for this goal, namely, the flow in a gradual expansion duct. The geometry of the duct depends on the value of the Reynolds number and the duct becomes longer and straighter as the Reynolds number increases. In this problem, the lower wall boundary of the duct, $y_1(x)$, depends on the Reynolds number in the following form:

$$\begin{aligned} &y_1(x) = -[1 + f(x)] \\ &f(x) = [\tanh(2) - \tanh(2 - 30x/\mathrm{Re})]/2, \quad 0 \le x \le x_{out} = \mathrm{Re}/3 \end{aligned} \tag{26}$$

where the symmetry plane is located at$y = 0$. The inlet boundary condition at $x = 0$ is given in terms of the Cartesian velocity components (u,v) as

$$\begin{aligned} &u = \frac{3}{2}u_0\left(1 - y^2\right) \\ &v = 0 \end{aligned} \tag{27}$$

The no-slip and impermeability conditions are imposed at the channel (see Fig. 16 for the problem schematic and the generated grid).

Fig. 17 presents the computed flow field shown by the streamlines for this test case with *Re* = 100. It is observed that the flow is separated and two vortices are created. The flow is reattached after some distance where the nozzle becomes straight. Fig. 18 shows a comparison of the CCSLBM results for the wall vorticity of the duct with different simulation results reported in [40] and also the results of Clitte et al. [41] that exhibits good agreement. The recirculation zone is accurately resolved by the CCSLBM solution which is indicated by the negative wall vorticity values.

7. Extension of the solution methodology to 3D

Here, the 2D formulation of the CCSLBM in the generalized curvilinear coordinates is extended to three-dimension. Now, the LB Eq. (2) is transformed into the 3-D generalized curvilinear coordinates in which the physical and computational planes are denoted by $(x, y, z)$and(ξ, η, ζ), respectively:

$$\begin{aligned} &\xi = \xi(x, y, z) \\ &\eta = \eta(x, y, z) \\ &\zeta = \zeta(x, y, z) \end{aligned} \tag{28}$$

where the relationship between the physical and computational domains satisfies the following condition

$$\begin{bmatrix} \xi_x & \eta_x & \zeta_x \\ \xi_y & \eta_y & \zeta_y \\ \xi_z & \eta_z & \zeta_z \end{bmatrix} = \begin{bmatrix} x_\xi & x_\eta & x_\zeta \\ y_\xi & y_\eta & y_\zeta \\ z_\xi & z_\eta & z_\zeta \end{bmatrix}^{-1} \tag{29}$$

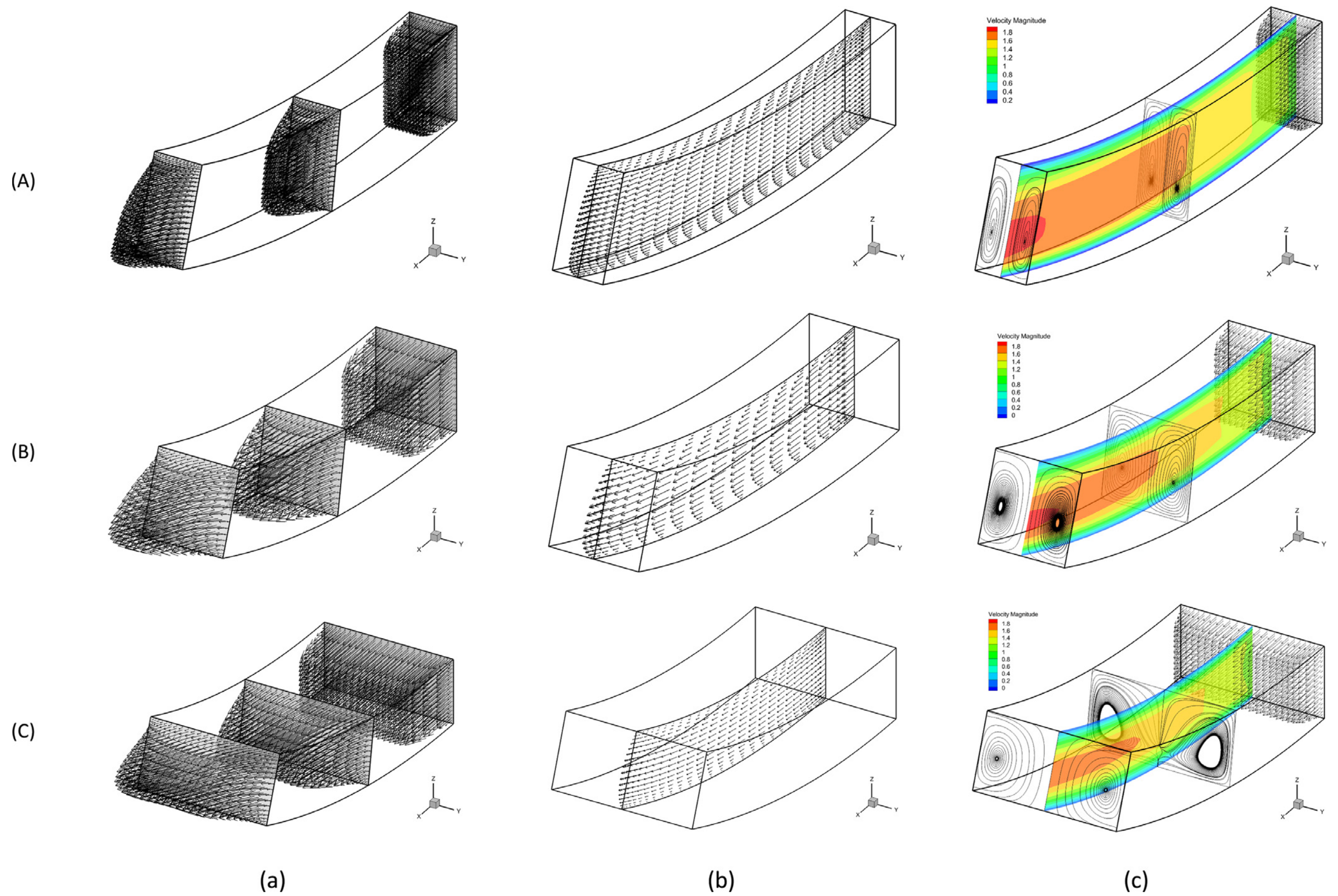


Fig. 26. Computed flow field shown by (a) velocity vectors at three cross sections, (b) velocity vectors at midplane, and (c) streamlines at three cross sections and velocity contours at midplane for the 3D duct problem for $\phi = 21^o$ and (A) $\gamma = 0.5$, (B) $\gamma = 1.0$, and (C) $\gamma = 2.0$.

and the Jacobian of the transformation $J$is denoted by

$$J = \begin{vmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial x}{\partial \eta} & \frac{\partial x}{\partial \zeta} \\ \frac{\partial y}{\partial \xi} & \frac{\partial y}{\partial \eta} & \frac{\partial y}{\partial \zeta} \\ \frac{\partial z}{\partial \xi} & \frac{\partial z}{\partial \eta} & \frac{\partial z}{\partial \zeta} \end{vmatrix} = x_\xi \left(y_\eta z_\zeta - y_\zeta z_\eta\right) + x_\eta \left(y_\zeta z_\xi - y_\xi z_\zeta\right) + x_\zeta \left(y_\xi z_\eta - y_\eta z_\xi\right) \quad (30)$$

In 3D, the discrete Boltzmann-BGK equation can be written as

$$\frac{\partial f_k}{\partial t} + \left(\tilde{e}_{k\xi}\frac{\partial f_k}{\partial \xi} + \tilde{e}_{k\eta}\frac{\partial f_k}{\partial \eta} + \tilde{e}_{k\zeta}\frac{\partial f_k}{\partial \zeta}\right) = -\frac{1}{\tau}(\, f_k - f_k^{eq}), \quad k = 0, 1, \ldots, 18 \quad (31)$$

where the subscript $k$denotes the direction of the particle speed. In the D3Q19 discrete Boltzmann model, the microscopic velocities (see Fig. 19) are given as [32]

$$\mathbf{e}_k = \begin{bmatrix} e_{kx} \\ e_{ky} \\ e_{kz} \end{bmatrix} = \begin{bmatrix} 0 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 & -1 & 0 & 0 & 1 & -1 & 0 & 0 & 1 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 & -1 & 0 & 0 & 1 & -1 & 1 & -1 & 1 & -1 \end{bmatrix} \quad (32)$$

where

$$\begin{aligned} \tilde{e}_{k\xi} &= e_{kx}\xi_x + e_{ky}\xi_y + e_{kz}\xi_z \\ \tilde{e}_{k\eta} &= e_{kx}\eta_x + e_{ky}\eta_y + e_{kz}\eta_z \\ \tilde{e}\zeta &= e_{kx}\zeta_x + e_{ky}\zeta_y + e_{kz}\zeta_z \end{aligned} \quad (33)$$

In this formulation, the equilibrium distribution function is defined as

$$f_k^{eq} = \alpha_k\Big(p + p_0\Big(3(\boldsymbol{e}_k \cdot \boldsymbol{u}) + \frac{9}{2}(\boldsymbol{e}_k \cdot\ \boldsymbol{u})^2 - \frac{3}{2}(\boldsymbol{u}\,.\,\boldsymbol{u})\Big)\Big) \quad (34)$$

where for the D3Q19 model

$$\begin{cases} \alpha_k = 2/36 & k = 0, 2, \ldots, 5 \\ \alpha_k = 1/36 & k = 6, 8, \ldots, 17 \\ \alpha_k = 12/36 & k = 18 \end{cases} \quad (35)$$

and $\boldsymbol{u} = (u, v, w)$ is the velocity vector.

The pressure p and the macroscopic velocity vector u are obtained from the following relations:

$$p = \sum_{k=0}^{18} f_k\,, \quad p_0\,\boldsymbol{u} = \sum_{k=0}^{18} \boldsymbol{e}_k\, f_k \quad (36)$$

The 3D LB Eq. (31) can be written in the following form

$$\frac{\partial f_k}{\partial t} = R_k \quad (37)$$

where

$$R_k = -\left(\tilde{e}_{k\xi}\frac{\partial f_k}{\partial \xi} + \tilde{e}_{k\eta}\frac{\partial f_k}{\partial \eta} + \tilde{e}_{k\zeta}\frac{\partial f_k}{\partial \zeta}\right) - \frac{1}{\tau}(f_k - f_k^{eq}) \quad (38)$$

Again, the spatial derivatives in the 3D LB equation are discretized by the Chebyshev collocation spectral method to obtain high accuracy solutions. Thus, the right hand-side of Eq. (37) can

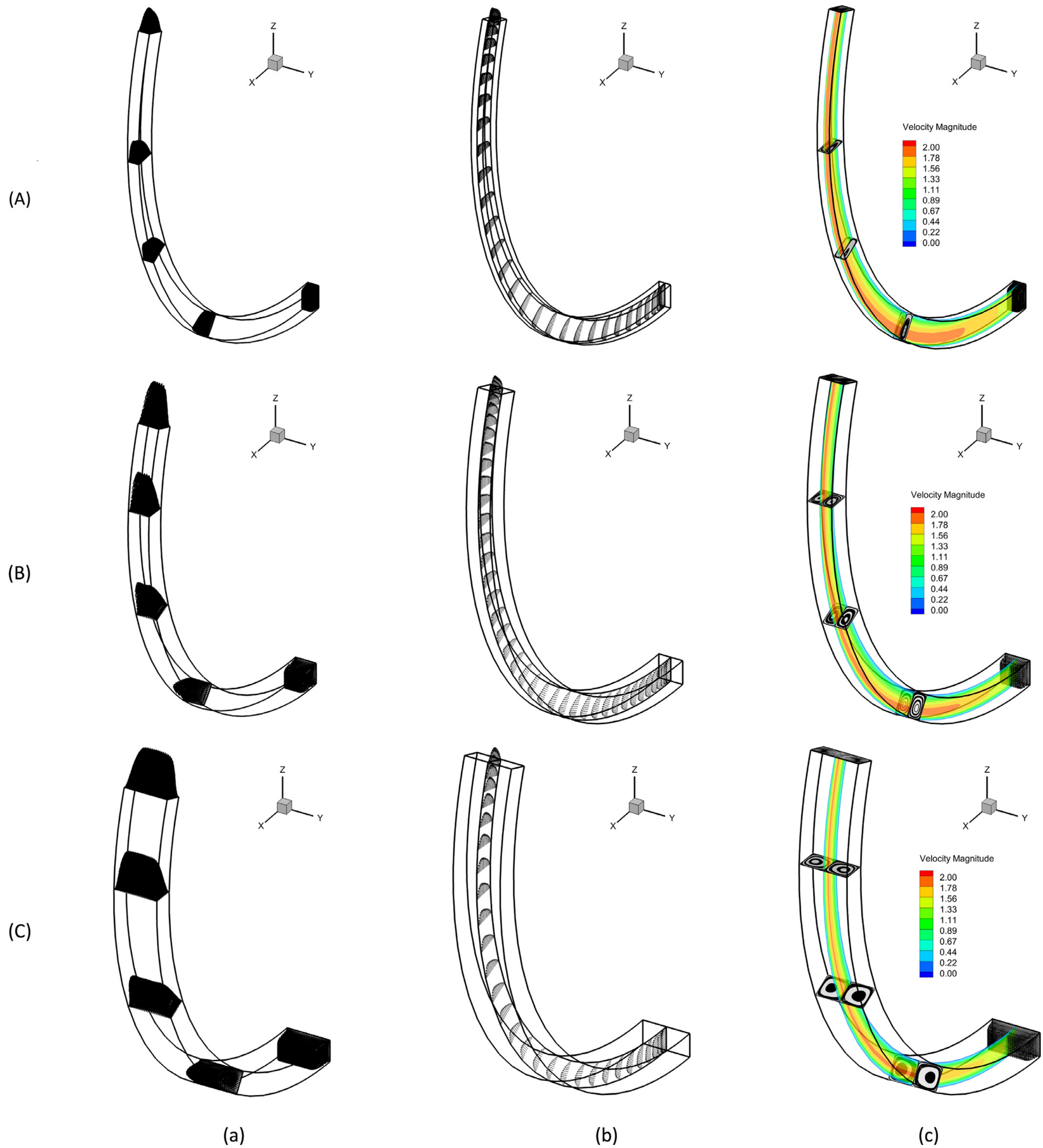


Fig. 27. Computed flow field shown by (a) velocity vectors at three cross sections, (b) velocity vectors at midplane, and (c) streamlines at three cross sections and velocity contours at midplane for the 3D duct problem for $\phi = 110^o$ and (A) $\gamma = 0.5$, (B) $\gamma = 1.0$, and (C) $\gamma = 2.0$.

be written as

$$R_{k_{m,n,l}} = -\left(\tilde{e}_{k\xi} \sum_{i=0}^{N_\xi} D_{\xi_{m,i}} f_{k_{i,n,l}} + \tilde{e}_{k\eta} \sum_{i=0}^{N_\eta} D^T_{\eta_{i,n}} f_{k_{m,i,l}} + \tilde{e}_{k\zeta} \sum_{i=0}^{N_\zeta} D^T_{\zeta_{i,l}} f_{k_{m,n,i}}\right) - \frac{1}{\tau}\left(f_{k_{m,n,l}} - f^{eq}_{k_{m,n,l}}\right) \quad (39)$$

where m, n and l indicate the number of grid points in the ξ-, η-, and ζ-direction, D_ξ indicates the derivative matrix in the ξ-direction, and D^T_η and D^T_η are the transpose of the collocation derivative matrices in the η- and ζ-direction, respectively. In addition, N_ξ, N_η, and N_ζ are the polynomial degrees in each direction. Similar to the 2D simulations, the temporal term in Eq. (37) is dis-

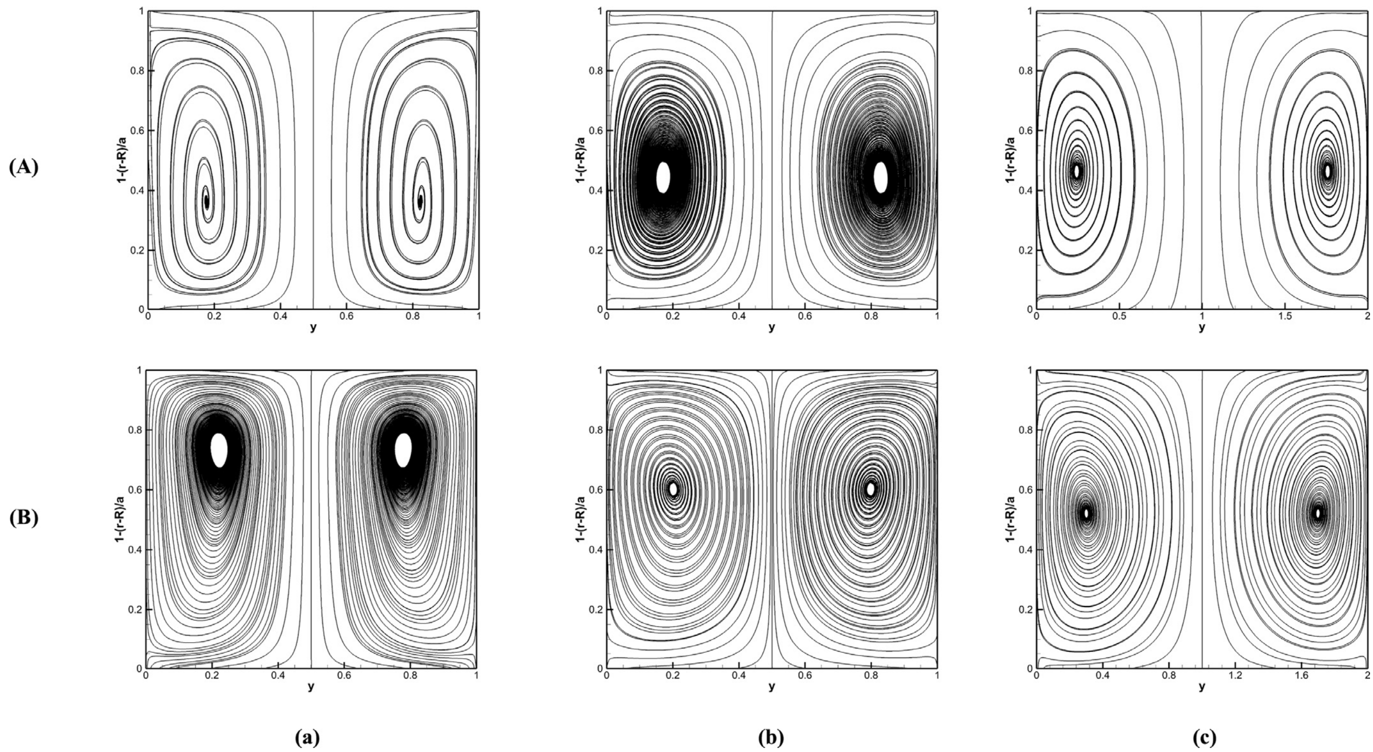


Fig. 28. Computed flow field shown by streamlines at outlet for the 3D curved duct problem for (A) $\phi = 21^{\circ}$ and (B) $\phi = 110^{\circ}$, and (a) $\gamma = 0.5$, (b) $\gamma = 1.0$, and (c) $\gamma = 2.0$.

cretized by the fourth-order Runge–Kutta scheme, not given here for brevity.

8. 3D numerical results

8.1. 3D flow in curved ducts of rectangular cross-sections duct

The study of viscous flow in curve ducts constitutes a problem of fundamental interest in the field of internal fluid mechanics. It serves as a model problem for understanding some of the important phenomena occurring in flows through turbo machinery blade passages, aircraft intakes, diffusers and heat exchangers.

The presence of longitudinal curvature generates centrifugal forces which act at right angles to the main flow direction and distort the flow. This effect can be analyzed in terms of the Dean number K and the duct curvature ratio R/D and therefore, an analysis for predicting three-dimensional viscous internal flows, even for simplified configurations and laminar incompressible cases, is of considerable interest. The Dean number is defined as $K = \mathrm{Re}\sqrt{D/R}$ where R is the radii of the center of rotation of the duct and $D$and Re are the hydraulic diameter of the duct cross section and the Reynolds number, respectively, as follows:

$$D = \frac{2ab}{a+b} \tag{40}$$

$$\mathrm{Re} = \frac{u_0 D}{\upsilon} \tag{41}$$

The geometry of the curved duct can be conveniently described in terms of the toroidal coordinate system (r, ϕ, y), as shown in Fig. 20(a). Fig. 20(b) illustrates different geometries that are investigated here to assess the accuracy of the CCSLBM in the simulation of the 3-D problems in the generalized curvilinear coordinate. The generated grids for the 3D curved duct problem for the case 1 for $\phi = 21^{o}$ and $110^{o}$are shown in Fig. 21. The Chebyshev–Gauss–Lobatto points are used in each direction.

For the curved ducts of rectangular cross-sections, the initial and boundary conditions at the initial plane, i.e., at $\phi = 0$, are given as follows:

$$v = w = 0,$$
$$u = u_0\left(\frac{494209}{419904}\right)\left\{\left[1-\left(\frac{2z-a}{a}\right)^{18}\right]^2\left[1-\left(\frac{2y-b}{b}\right)^{18}\right]^2\right\} \tag{42}$$

where u is defined such that the average velocity magnitude in the inlet boundary becomes $\bar{u} = u_0$, as illustrated in Fig. 22. Note that the spectral methods have problem with discontinues variations, and one should use a smooth velocity profile at the inlet boundary. Therefore, the proposed velocity profile u given in Eq. (42) can be used for the internal flow simulations and it will be seen that the numerical results based on this velocity profile are comparable with those of the uniform velocity distribution $\bar{u} = u_0$ used by Ghia and Sokhey [42].

Results are obtained for the curved ducts of rectangular cross-sections for different values of the problem parameters. These consist mainly of the aspect ratio $\gamma = b/a$ for the rectangular cross-section and the curvature ratio R/D as

$$\begin{aligned} &\text{Case 1}: \ R/D = 21.0, \quad \gamma = 0.5 \\ &\text{Case 2}: \ R/D = 14.0, \quad \gamma = 1.0 \\ &\text{Case 3}: \ R/D = 10.5, \quad \gamma = 2.0 \end{aligned} \tag{43}$$

The effect of the variation of the aspect ratio on the development of the axial velocity profiles for these cases with $K = 55$ is shown in Figs. 23 and 24 for $\phi = 110^{o}$ and 21^{o}, respectively. The corresponding velocity profiles along the midlines at the outlet plane are shown for three values of the aspect ratio $\gamma = 0.5$, 1.0, and 2.0. From these figures, it is indicated that a decrease in the aspect ratio causes an increase in the peak velocity with a simultaneous outward shift in the location of this peak value and it is due to the centrifugal forces. The present results by applying the CCSLBM are comparable with those of Ghia and Sokhey [44] by solving the Navier–Stokes equations. The convergence history of the solution based on the L_2-norm error is shown in Fig. 25. Also, the velocity field in different cross sections of the duct are illustrated in Figs. 26 and 27. Again, it is observed that the centrifugal forces cause an outward shift in the peak location of the velocity profiles across the curved duct. Therefore, the maximum velocity is located close to the lower wall of the duct and this is obvious in Figs. 26(b) and 27(b) where the velocity vectors in the midplane of the duct are shown. The complex flow structure of this problem can be seen properly in this figure.

Fig. 28 illustrates the computed outlet streamlines of the different ducts. It is observed that by increasing γ from 0.5 to 2.0 at both $\phi = 110^o$and 21^o, the locations of the center of the vortices tends toward the center of the outlet plane. Note that the center of the vortices for $\phi = 21^o$ is located near the lower wall whereas for $\phi = 110^o$ it is located near the upper wall. As mentioned before, the center of the vortices for both $\phi = 21^o$ and 110^o is moved toward to the center of the plane by increasing γ.

9. Conclusion

In this work, the Chebyshev collocation spectral lattice Boltzmann method (CCSLBM) is implemented in the generalized curvilinear coordinates to develop an accurate and efficient low-speed LB-based flow solver to be capable of handling curved geometries with non-uniform grids. The low-speed form of the LB equation with the Bhatnagar–Gross–Krook (BGK) approximation is transformed into the generalized curvilinear coordinates and then the spatial derivatives in the resulting equation are discretized by using the Chebyshev collocation spectral scheme and the temporal term is discretized with the fourth-order Runge–Kutta scheme. The calculations are performed for different steady and unsteady low-speed flow benchmark problems to demonstrate the accuracy and robustness of the Chebyshev collocation spectral LBM (CCSLBM) applied. Some conclusions and remarks regarding the present study are as follows:

1- The computations indicate that the solution methodology proposed based on implementing the Chebyshev collocation spectral LBM (CCSLBM) in the generalized curvilinear coordinates is stable and robust for the problems simulated and the solution algorithm does not need any filtering or numerical dissipation for stability considerations. It is shown that the computed results obtained by the CCSLBM are in good agreement with the analytical solutions and also the available numerical results for the test cases considered.

2- An appropriate procedure for implementing the boundary conditions in the generalized curvilinear coordinates for the solution of the CCSLBM is proposed and applied. It is indicated that the physical boundary conditions applied by solving the governing equations in the generalized curvilinear coordinates on the boundaries based on given/calculated macroscopic values of the flow variables can resolve the flow field near the boundaries accurately.

3- The computational efficiency of the proposed solution methodology based on the Chebyshev collocation spectral LBM (CCSLBM) in the generalized curvilinear coordinates are examined by comparison with the second-order central finite-difference LBM (FDLBM) developed. It is demonstrated that the CCSLBM implemented in the generalized curvilinear coordinates provides more accurate and efficient solutions than the FDLBM in terms of the CPU time and memory usage and when a high accuracy solution is required the performance of the CCSLBM is more highlighted. Indications are that high accuracy solutions obtained by implementing the CCSLBM can be used as benchmark solutions for the assessment of other LBM-based flow solvers.

4- Results obtained by applying the Chebyshev collocation spectral LBM in the generalized curvilinear coordinates are also comparable with those of Navier–Stokes solvers. Note that the formulation of the Chebyshev collocation spectral LBM is simpler to implement and it can be considered as an appropriate alternative computational technique to Chebyshev collocation spectral Navier–Stokes solvers for precisely studying physical phenomena and simulating fluid dynamics over practical geometries.

5- The present solution methodology based on the implementation of the CCSLBM in the generalized curvilinear coordinates is given in both 2D and 3D. Note that most spectral based flow solvers in the literature have been developed in two-dimensions and the extension/implementation of these methods in three-dimensions in the generalized curvilinear coordinates for accurate and efficient simulating the flow over practical geometries can be valuable. Such an extension/implementation is performed in the present study. It is demonstrated that the CCSLBM is able to accurately and efficiently solve steady and unsteady low speed flows over 2D and 3D geometries.

6- The CCSLBM applied in the generalized curvilinear coordinates in the present study is based on the single domain technique. To further improve the efficiency and flexibility of the CCSLBM in computing the flow over more practical problems, one can apply the multidomain technique to the CCSLBM which is our future goal. The implementation of the multidomain technique to the CCSLBM can reduce the data dependencies resulting a better scalability of the solution method used in parallel and high performance computing.

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